Experimental Observation of Oceanic Forced Convection

Max Coppin, Bruno Deremble, Joel Sommeria

20 Mar 2025

Parameterizations for global dynamical models in Climatology, Astrophysics and Planetology

LEGI
LABORATOIRE DES ECOULEMENTS GEOPHYSIQUES ET INDUSTRIELS

IGE
Institut des géosciences de l'environnement

Outline

1. Introduction
2. Wind-driven ML deepening in ocean
3. Forced convection experiment
4. Longer time behavior of the ML
5. Conclusion



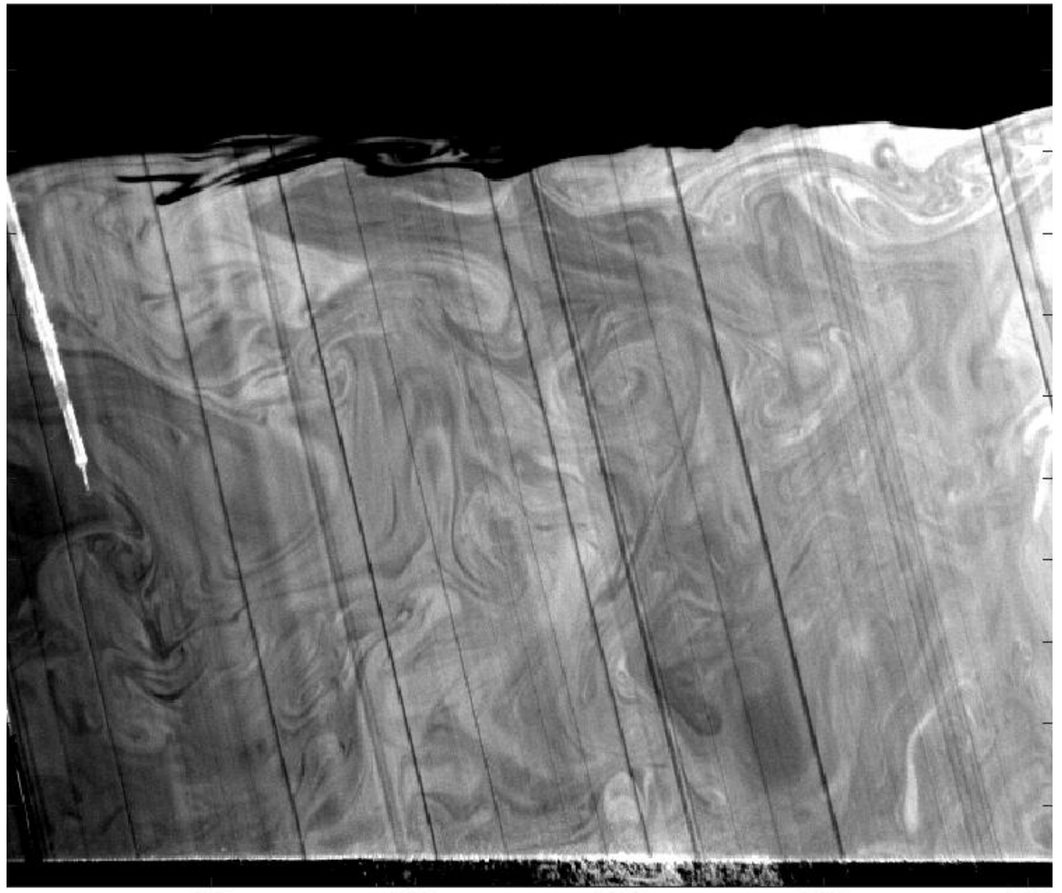
Credit : Facebook/Rachel Gordon

Introduction: Definition of Convection

Vertical motion of a fluid parcel caused by:

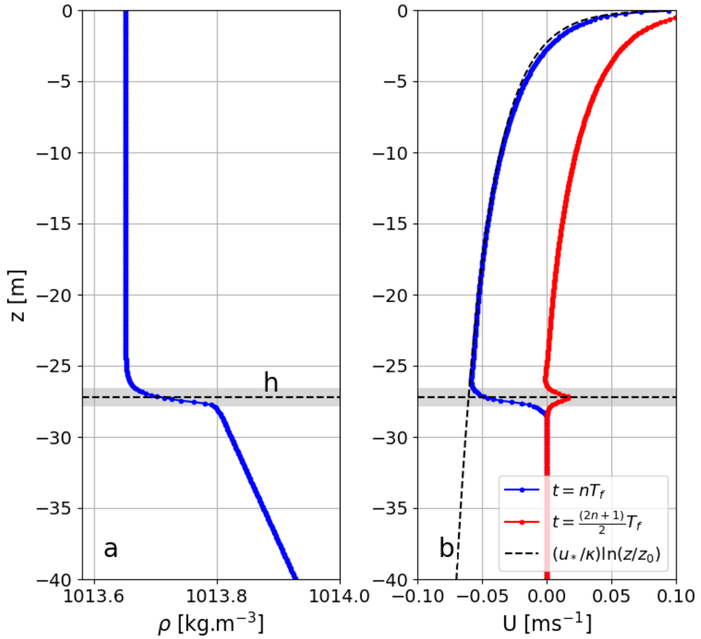
- **Buoyancy Force** → **Free convection**
- **External Force** → **Forced convection**

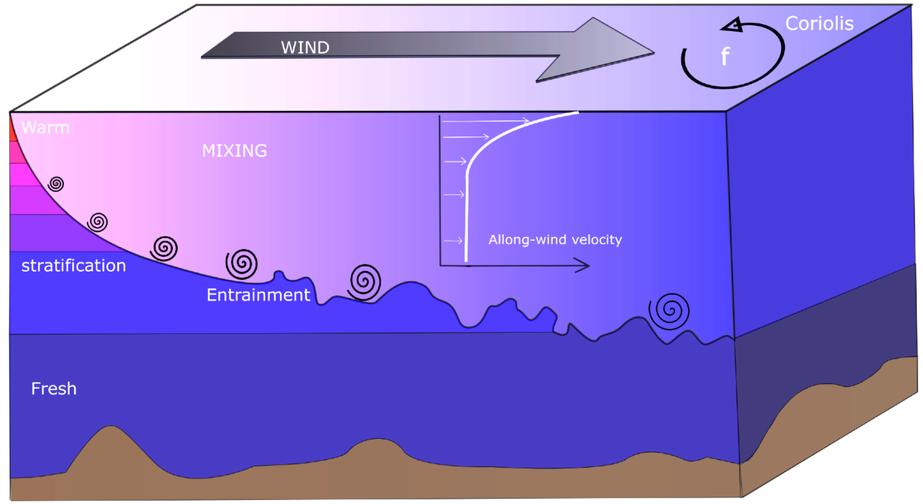




Wind-driven Oceanic Convection : Processes

Shear stress at the surface



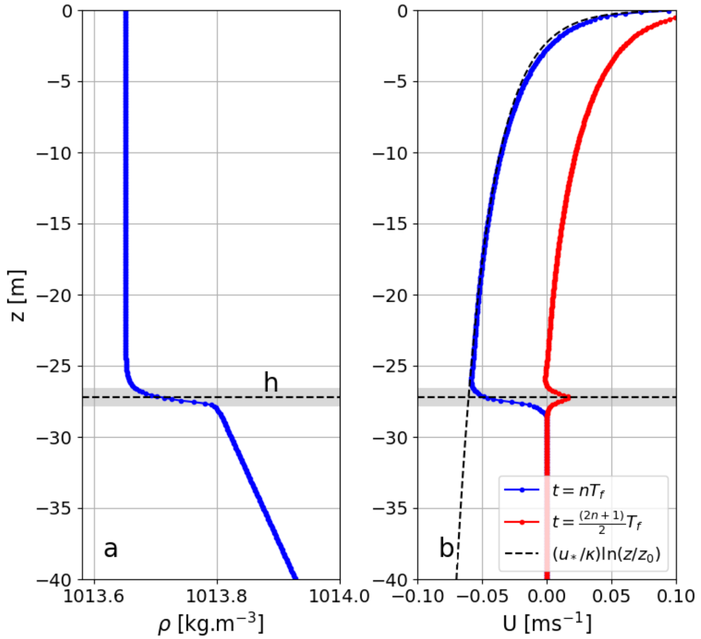


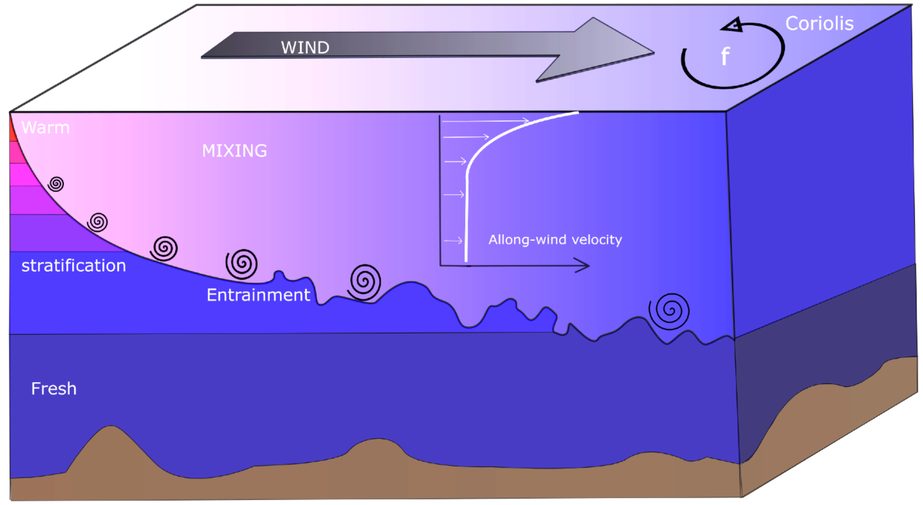
Forced Convection (Momentum Flux)

Wind-driven Oceanic Convection : Processes

Shear stress at the surface

Log layer / Ekman Layer





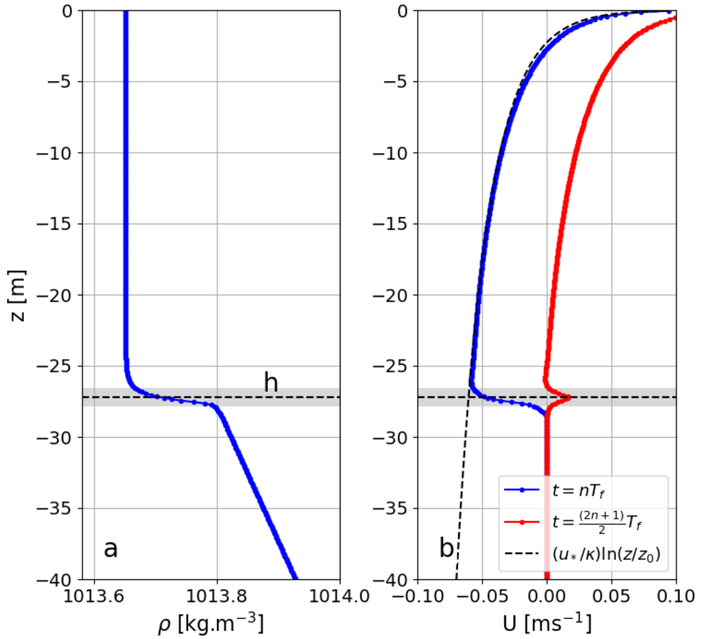
Forced Convection (Momentum Flux)

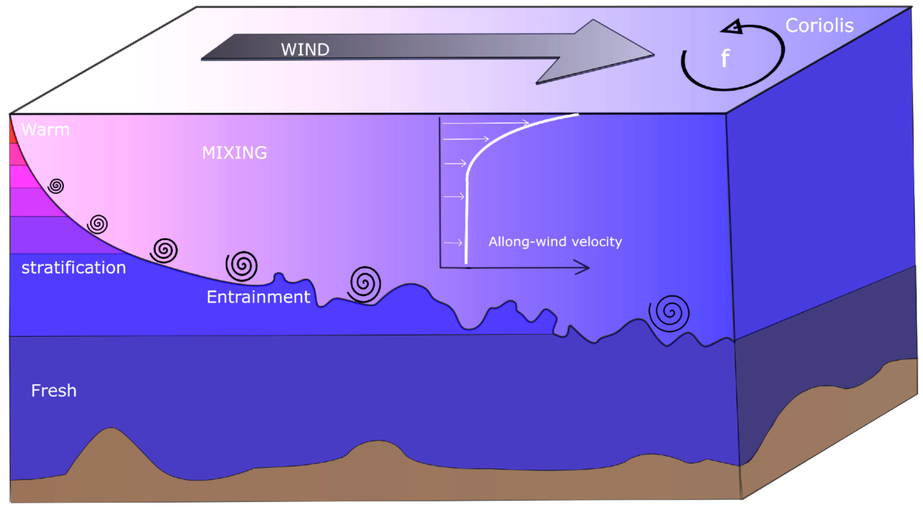
Wind-driven Oceanic Convection : Processes

Shear stress at the surface

Log layer / Ekman Layer

Entrainement turbulence





Forced Convection (Momentum Flux)

Wind-driven Oceanic Convection : Processes

Shear stress at the surface

↓

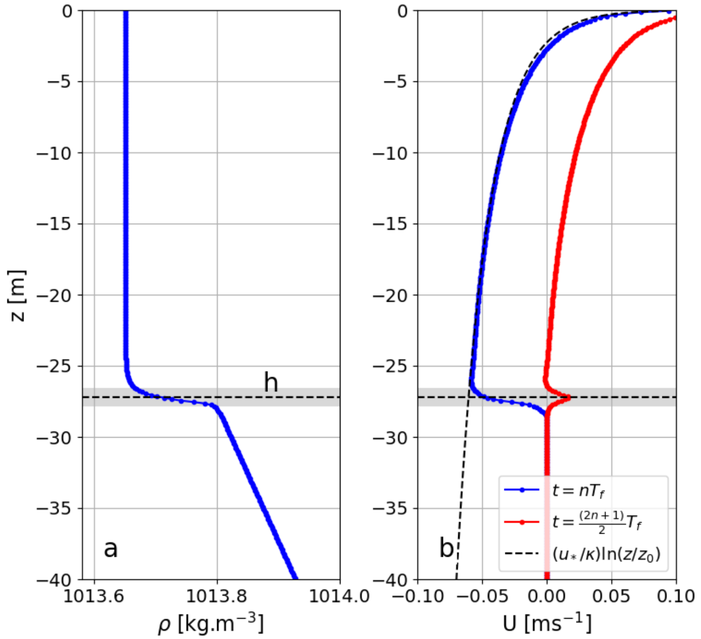
Log layer / Ekman Layer

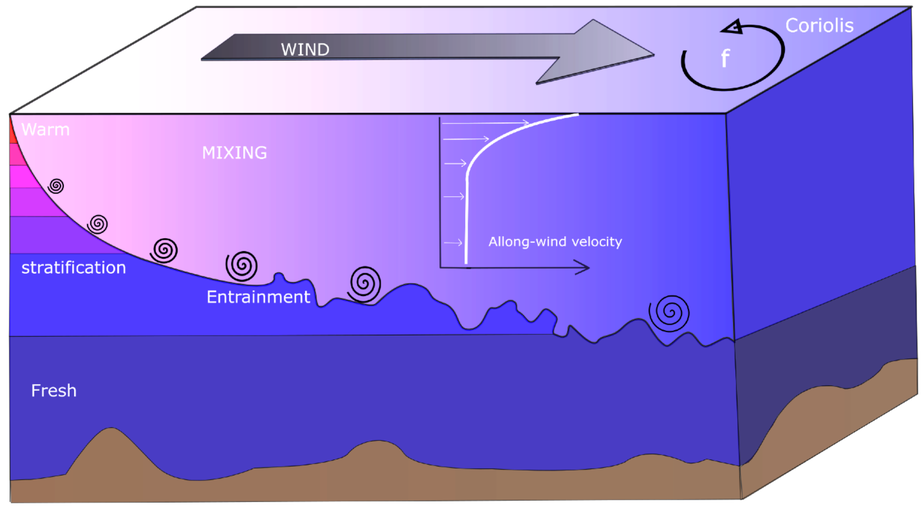
↓

Entrainement turbulence

↓

Erosion of the Stratification





Forced Convection (Momentum Flux)

Wind-driven Oceanic Convection: Scaling law

Pollard et al, 1973: Assumption

- Linear stratification
- Bulk consideration
- The entire stress at the bottom of ML deepen it
- Excess of energy by wind + rate of change of TKE + Dissipation = 0

$$u_{surface} > u_{bulk}$$

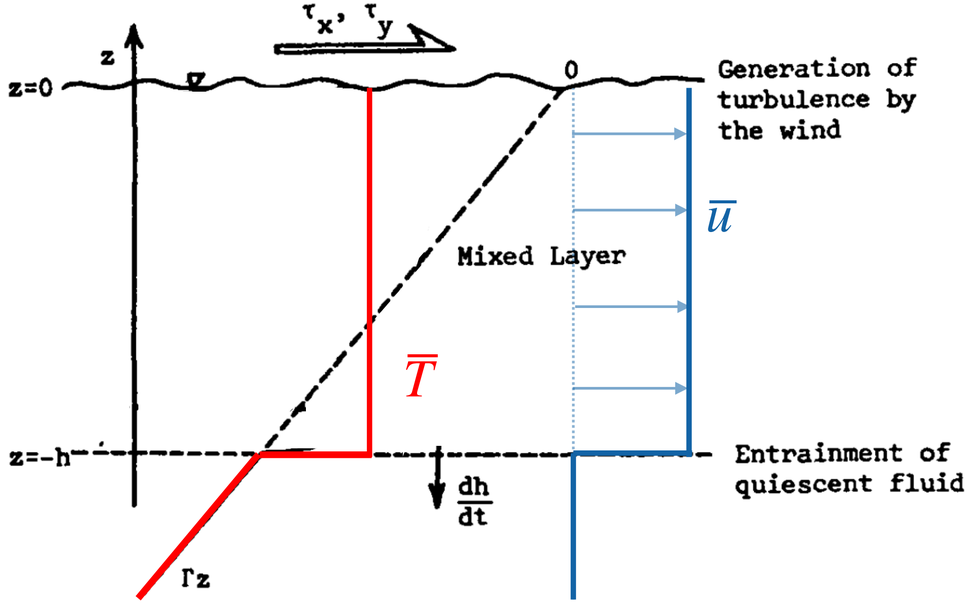


Figure modified from Cushman-Roisin, 1981

Pollard, R.T., Rhines, P.B., Thompson, R., 1973. The deepening of the wind-Mixed layer. Geophysical Fluid Dynamics.

Wind-driven Oceanic Convection: Scaling law

Pollard et al, 1973: Assumption

- Linear stratification
- Bulk consideration
- The entire stress at the bottom of ML deepen it
- Excess of energy by wind + rate of change of TKE + Dissipation = 0

$$u_{surface} > u_{bulk}$$

ML deepening scaling law

- Momentum of mixed Layer increase : $uh \propto u_*^2 t$
- Marginal stability state $\Rightarrow$ $Ri = \dfrac{h^2 N_0^2}{u^2} = Ri_c$

$$h_{(t)} \sim 2^{1/4} u_* \sqrt{\frac{t}{N_0}}$$

$$h_{(t)} = \frac{u_*}{\sqrt{N_0 f}} \left[4(1 - \cos(ft)\right]^{1/4}$$

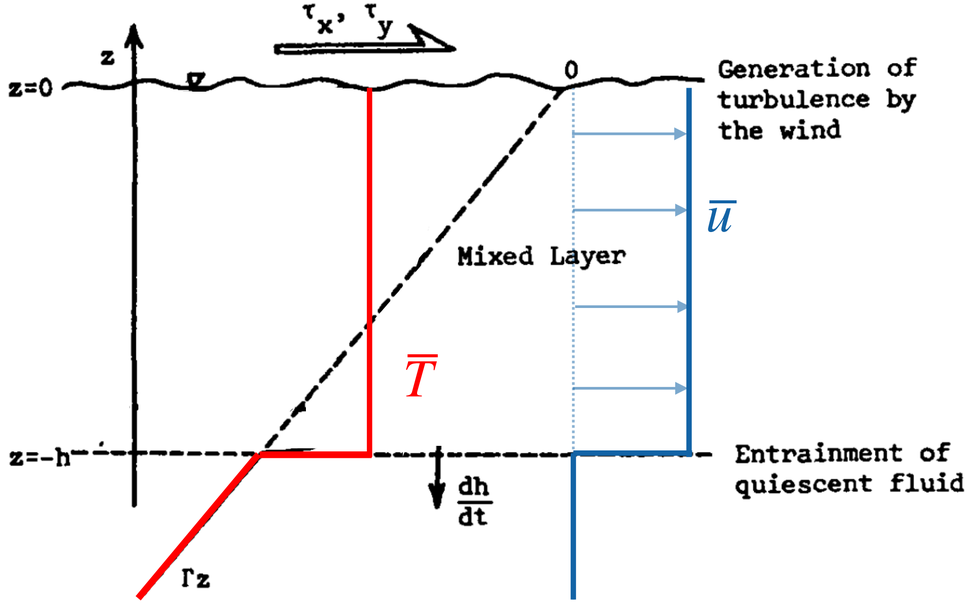


Figure modified from Cushman-Roisin, 1981

Pollard, R.T., Rhines, P.B., Thompson, R., 1973. The deepening of the wind-Mixed layer. Geophysical Fluid Dynamics.

Seminal Experiments

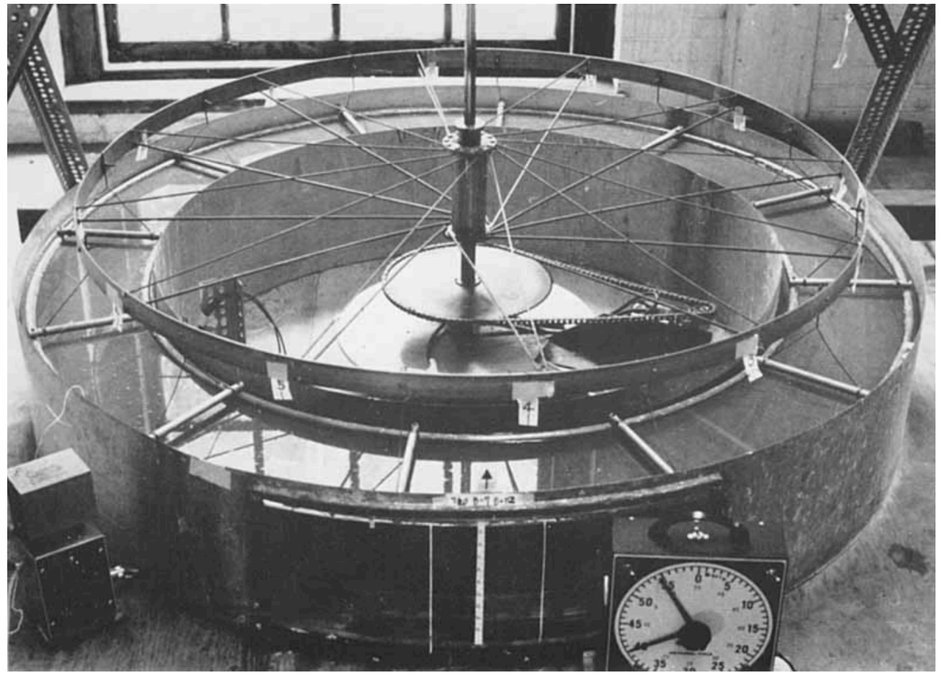


FIGURE 1. The experimental apparatus.

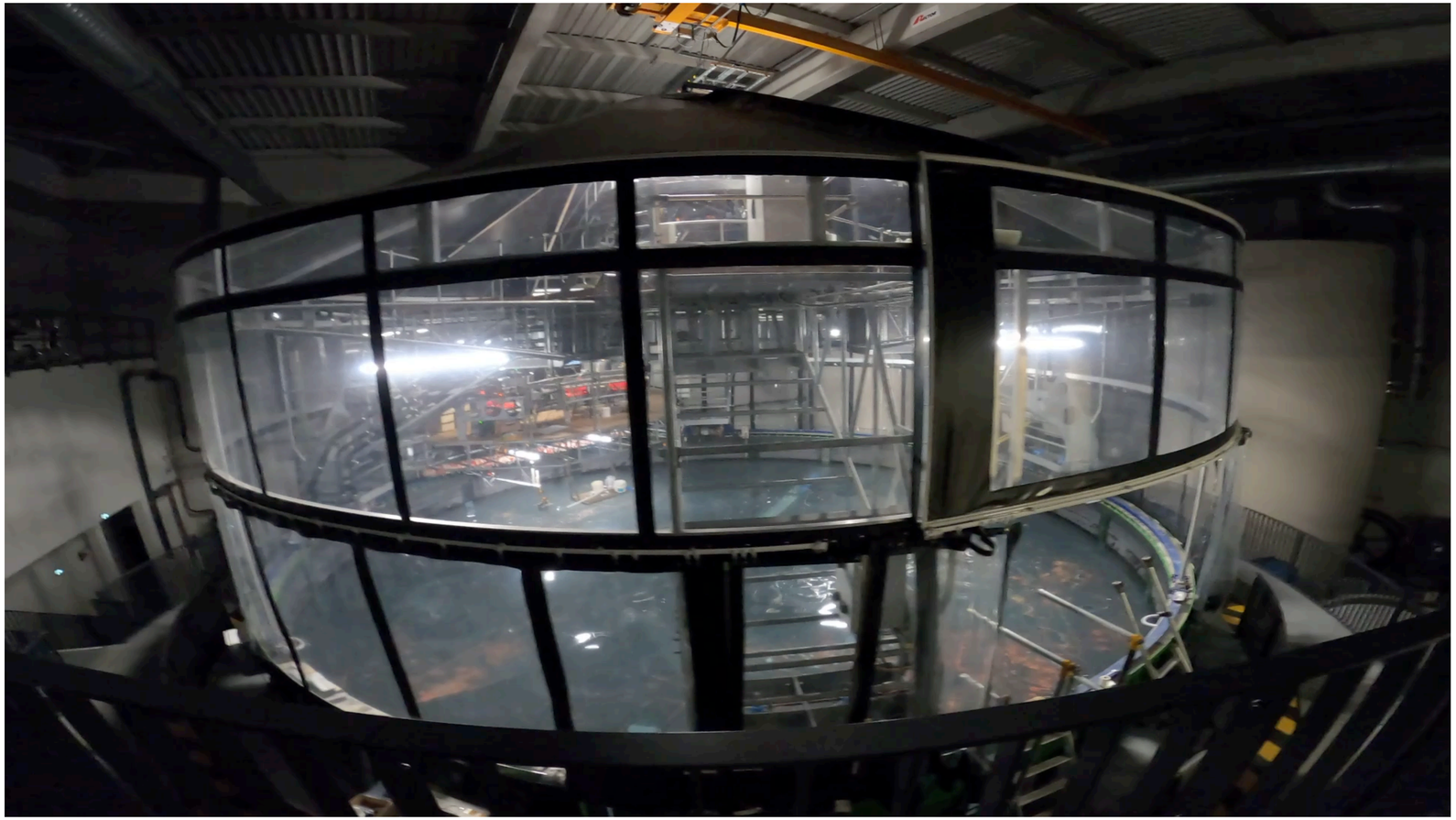
Kato - Philips 1969

- Mixed layer deepening rate : $h \sim t^{1/3}$
- Entrainment law : $E(Ri) = \dfrac{dh/dt}{u_*} = 2.5\, Ri^{-1}$

- They did not considered rotational effects
- The torque applied was tuned « by hand »
- No direct measurement of the density
- No direct measurement of the velocity

Kato, H., Phillips, O.M., 1969. On the penetration of a turbulent layer into stratified fluid. Journal of Fluid Mechanics 37, 643-655

Coriolis Platform

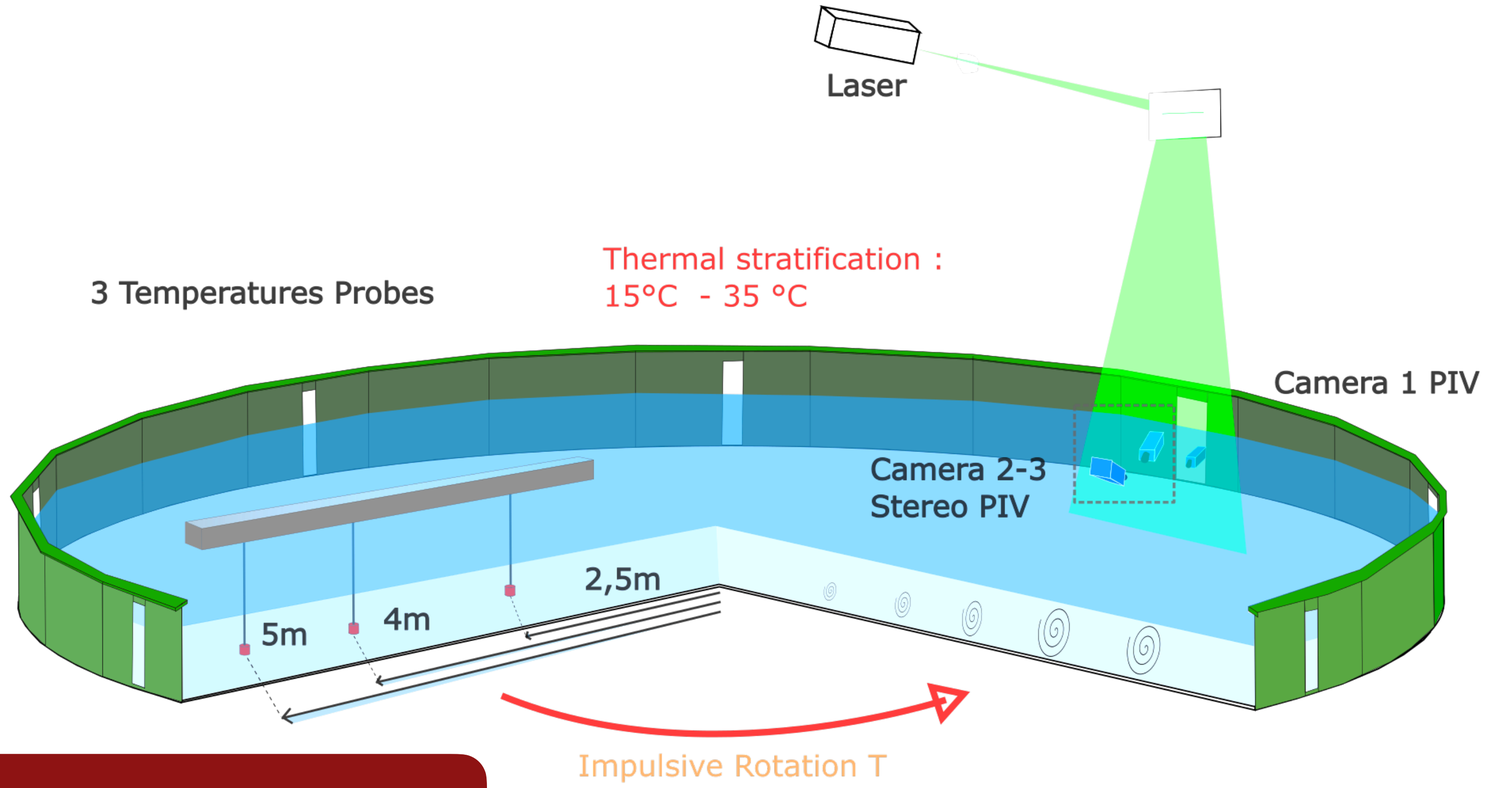


- **Diameter:** $13\,m$
- **Weight :** 350 Tones at full load
- **Maximum Speed:** $6\,rpm$
- **Max water height:** $1\,m$
- **Volume:** $132\,m^3$

- **Rossby Number** $Ro = \frac{U}{fL}$
- **Froude Number :** $Fr = \frac{U}{NL}$
- **Reynold Number:** $Re = \frac{UL}{\nu}$

Forced Convection Experiment: Apparatus

- **Acceleration of rotation (Spin-Up)**
- **Temperature stratification**
- **Temperature probes**
  - 3 Vertical profilers
- **Vertical laser sheet (30x25)cm**
  - PIV Stereo (2D - 3 components)



Control parameters

Friction : u_* **Rotation :** f **Stratification** $N^2 \equiv (\Delta T)$

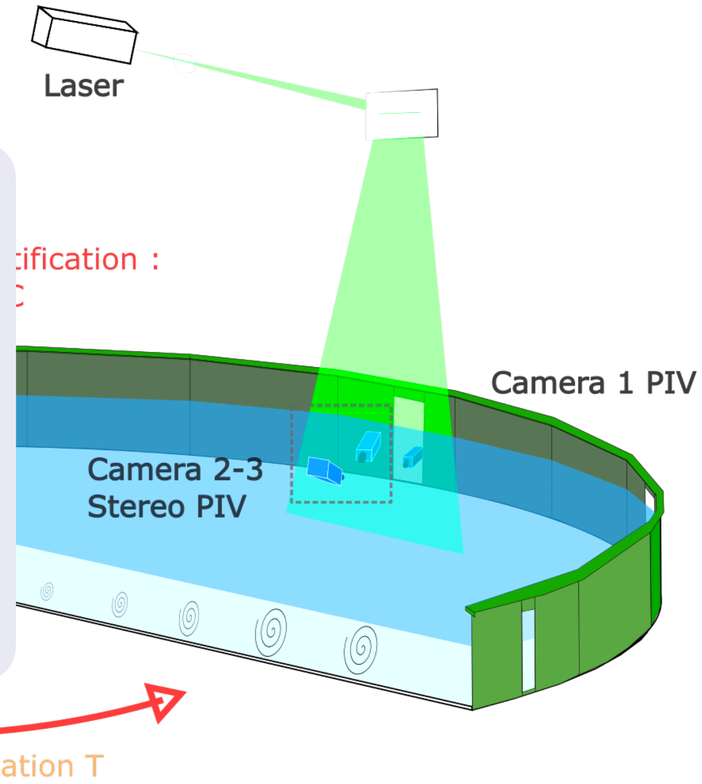
Forced Convection Experiment: Apparatus

- **Acceleration of rotation (Sp**
- **Temperature stratification**
- **Temperature probes**
  - 3 Vertical profilers
- **Vertical laser sheet (30x25)**
  - PIV Stereo (2D - 3 componen

All Scalings depends on this velocity friction

↓

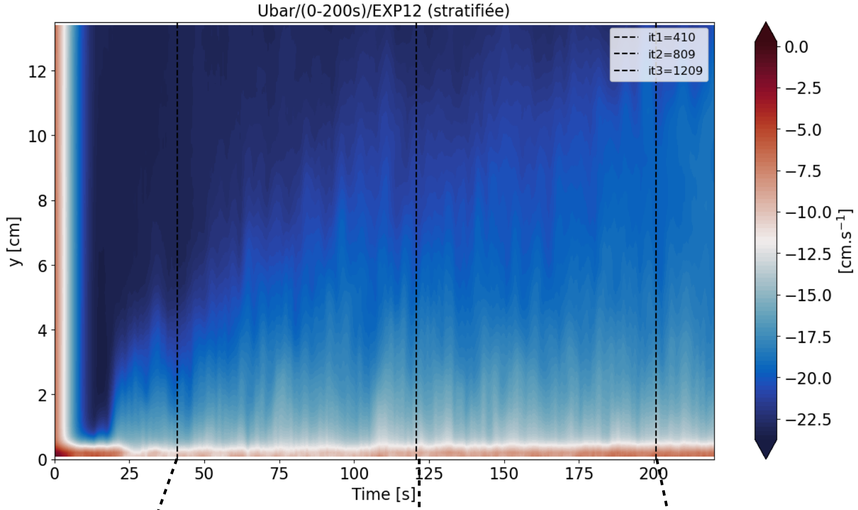
A proper characterization is needed

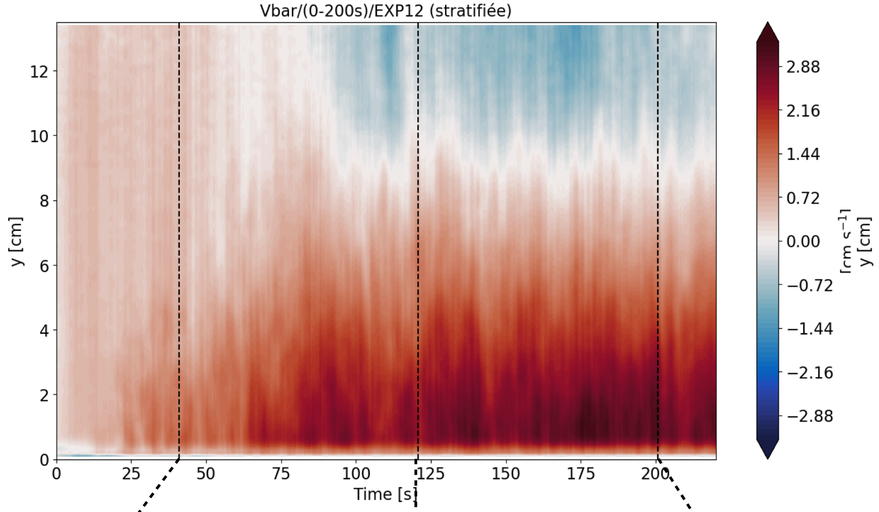


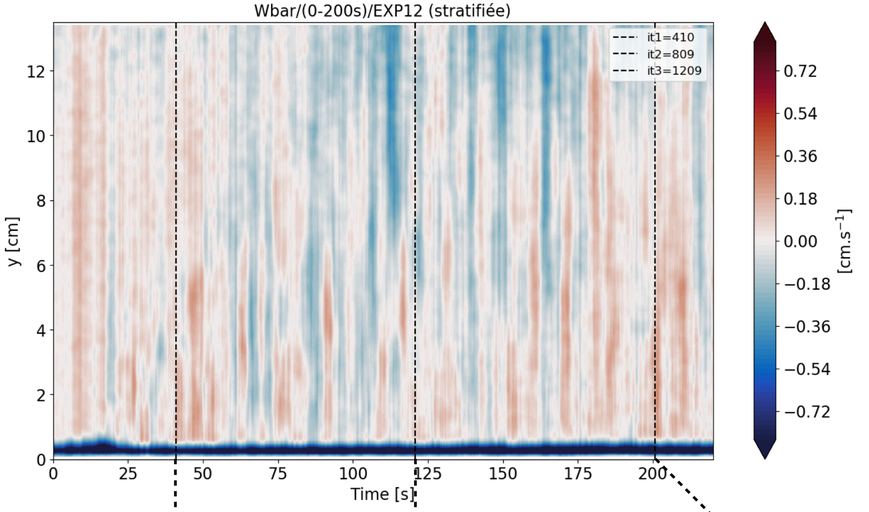
Control parameters

Friction : $u_* \sim 2\pi R\Omega/20$ **Rotation :** f **Stratification** $N^2 \equiv (\Delta T)$

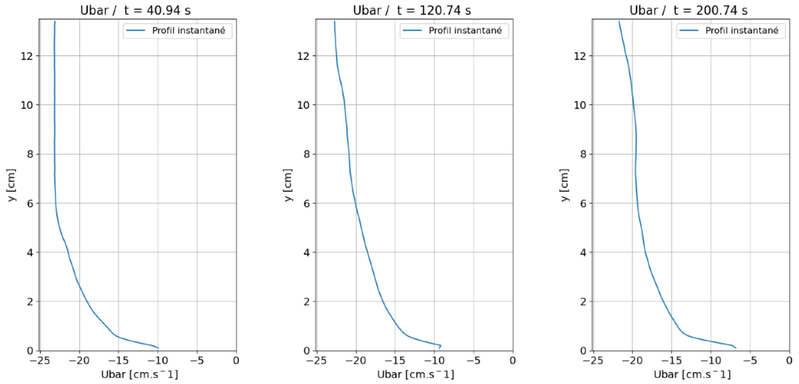
Forced Convection Experiment: Vertical profiles of velocity



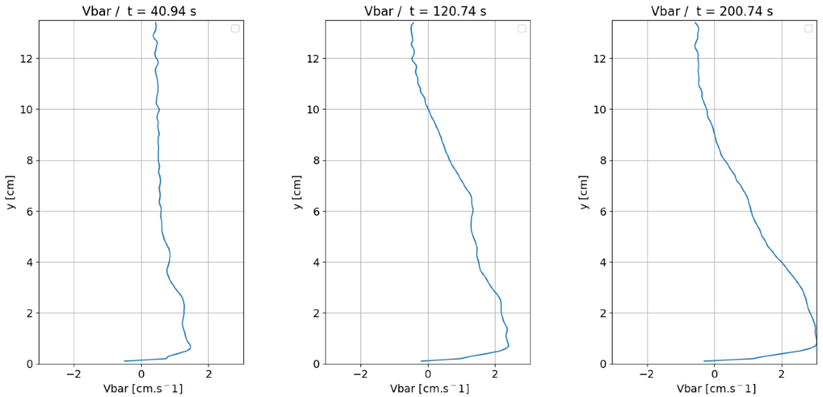




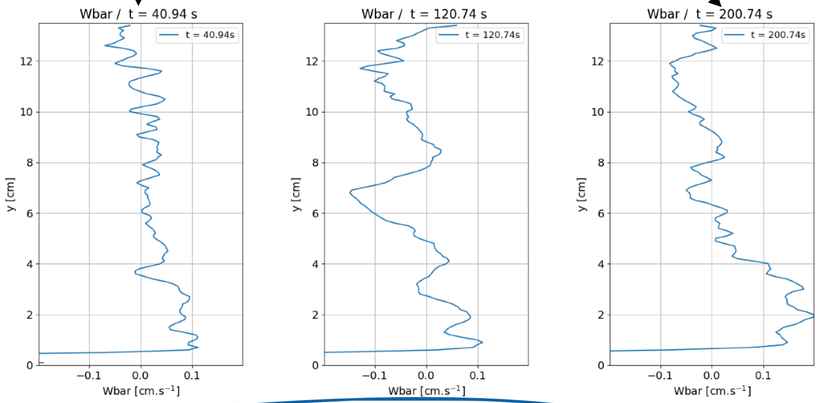
Profils verticaux de Ubar à différents instants



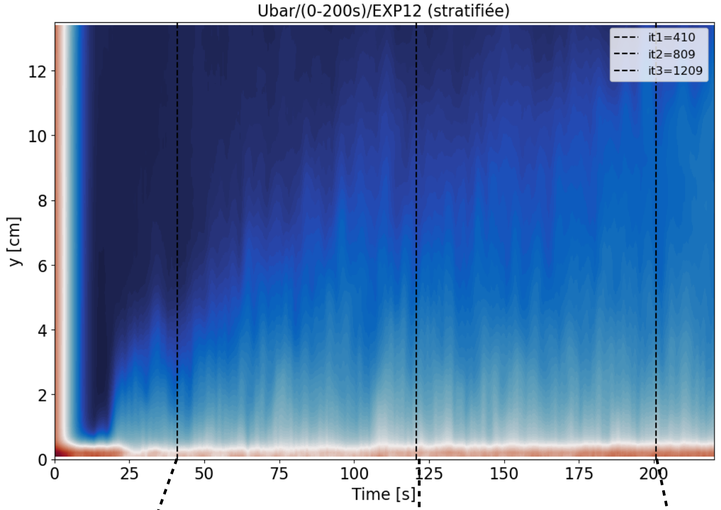
Profils verticaux de Vbar à différents instants

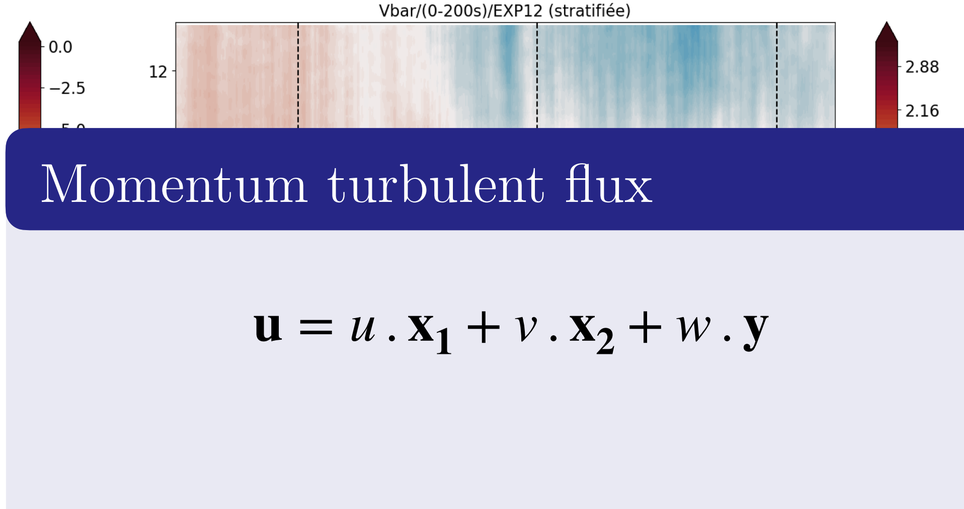


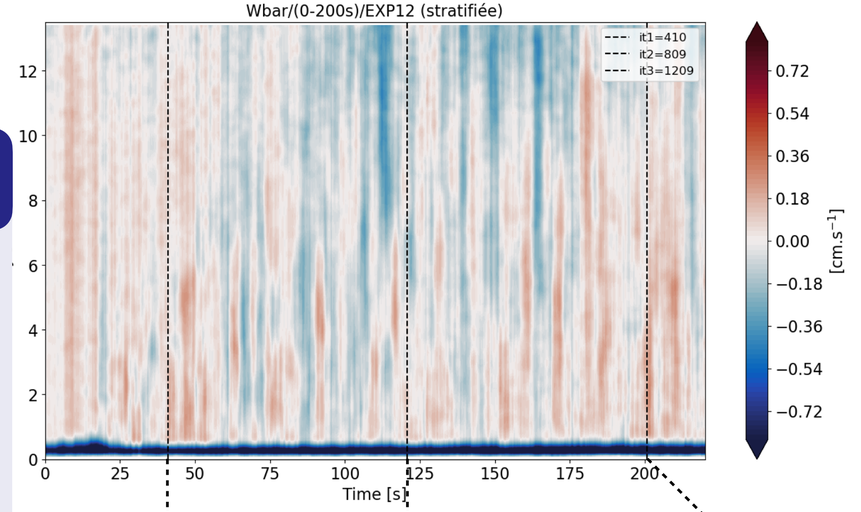
Profils verticaux de Wbar à différents instants



Forced Convection Experiment: Vertical profiles of velocity

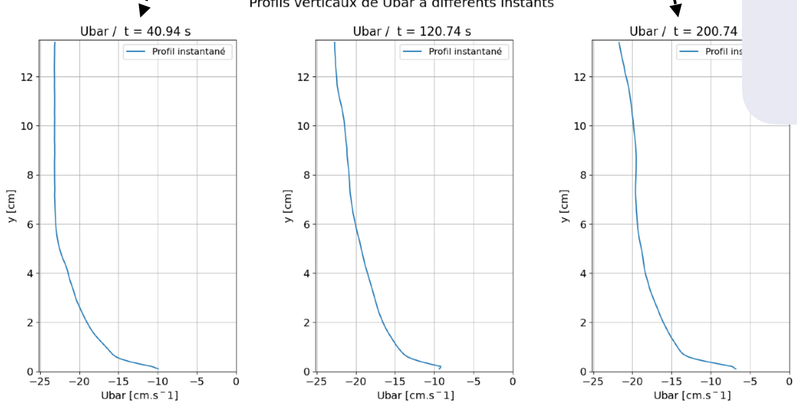


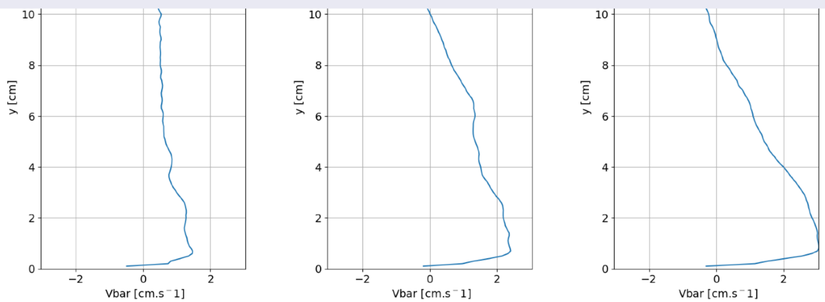


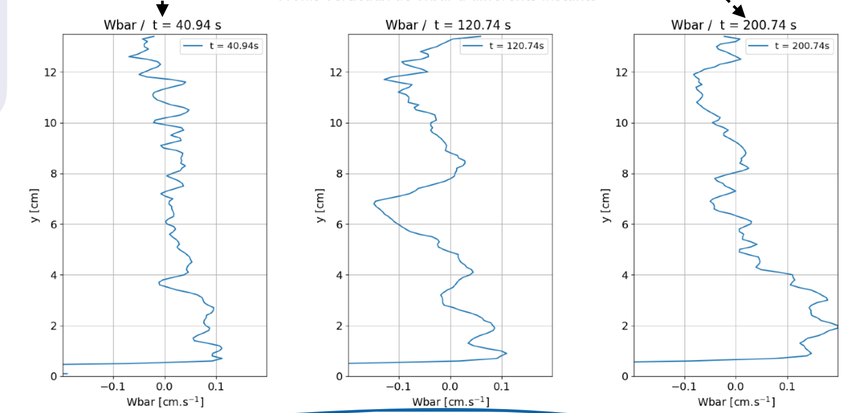


Momentum turbulent flux

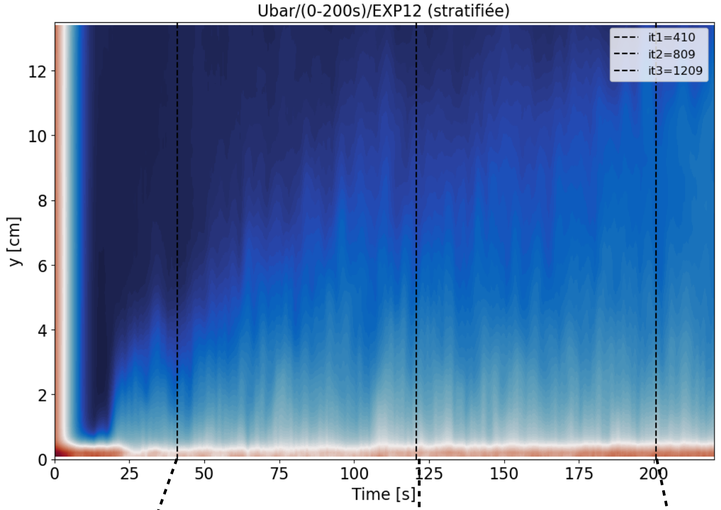
$$\mathbf{u} = u\,.\,\mathbf{x_1} + v\,.\,\mathbf{x_2} + w\,.\,\mathbf{y}$$

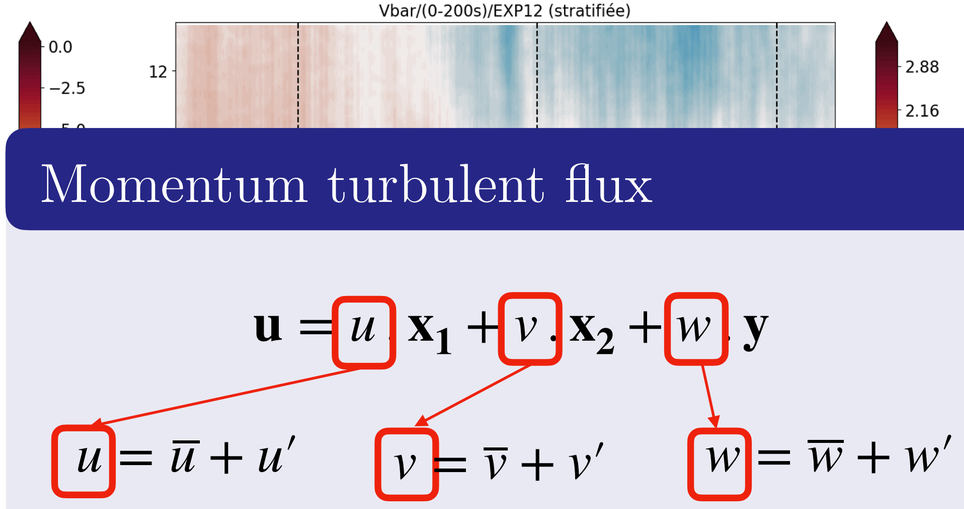


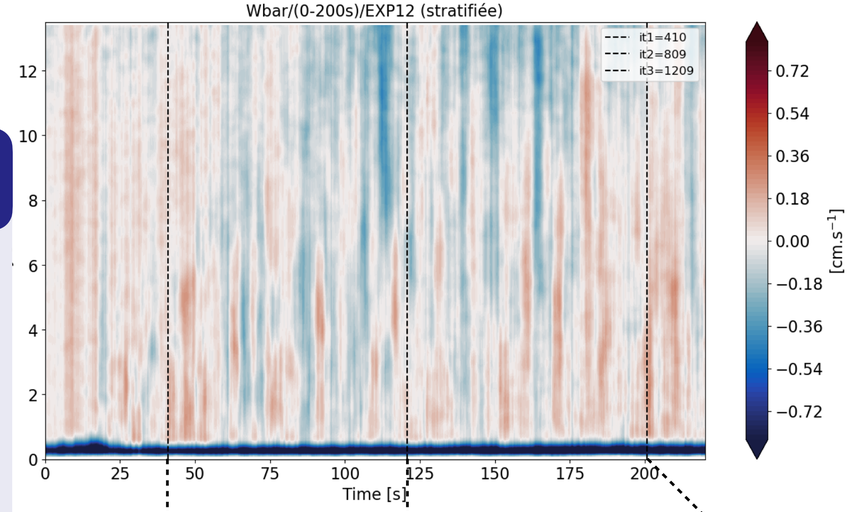




Forced Convection Experiment: Vertical profiles of velocity





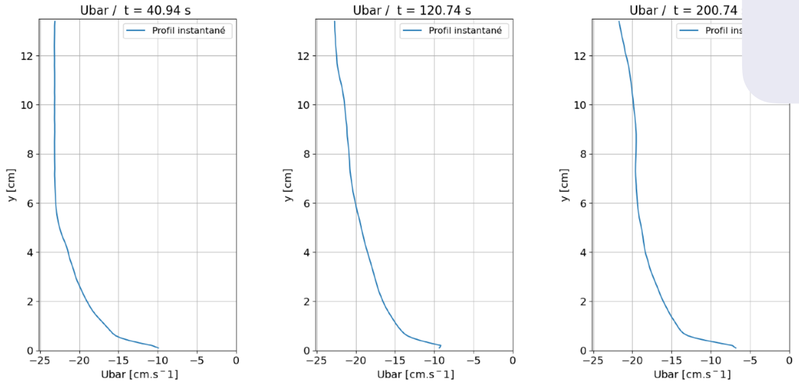


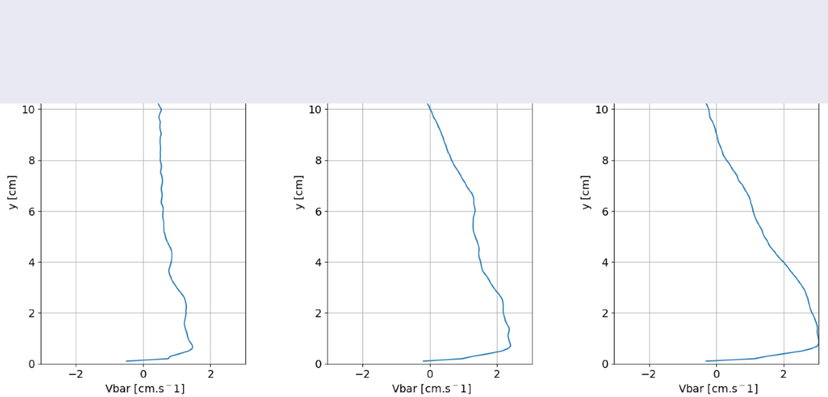
Momentum turbulent flux

$$\mathbf{u} = u\,.\,\mathbf{x_1} + v\,.\,\mathbf{x_2} + w\,.\,\mathbf{y}$$

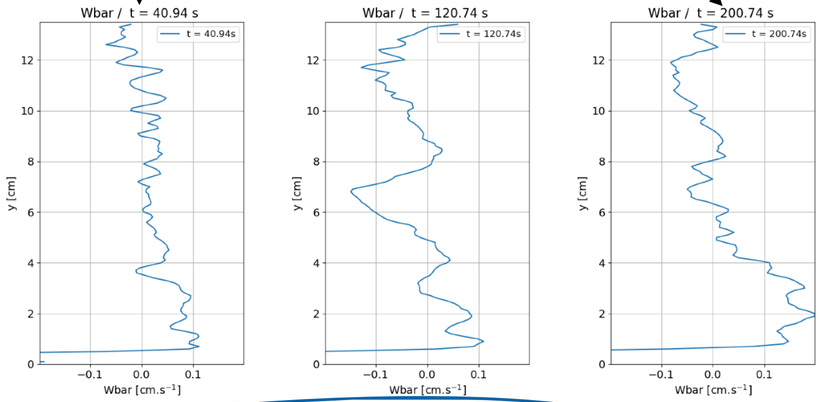
$$u = \overline{u} + u' \qquad v = \overline{v} + v' \qquad w = \overline{w} + w'$$

Profils verticaux de Ubar à différents instants

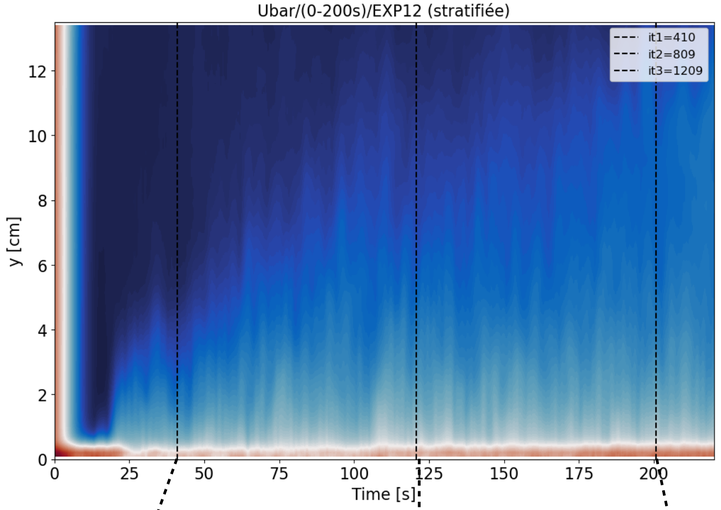


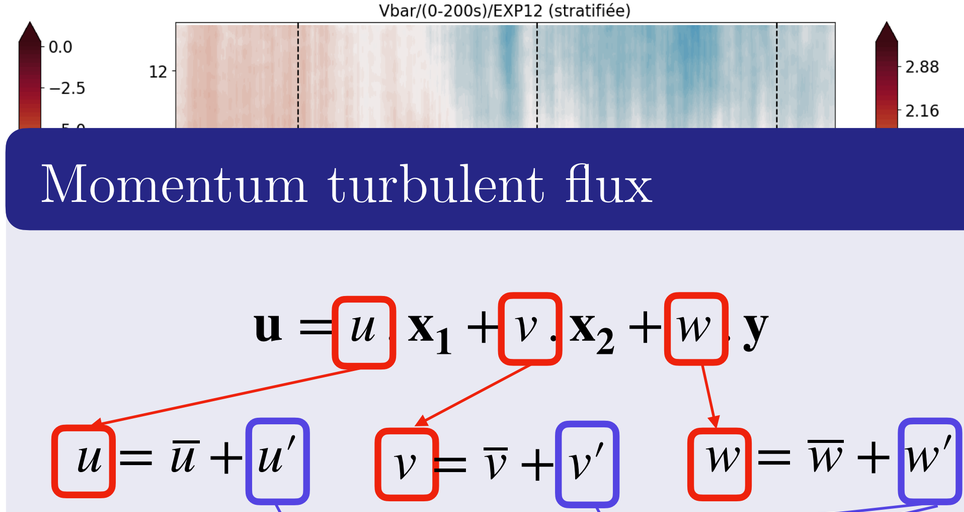


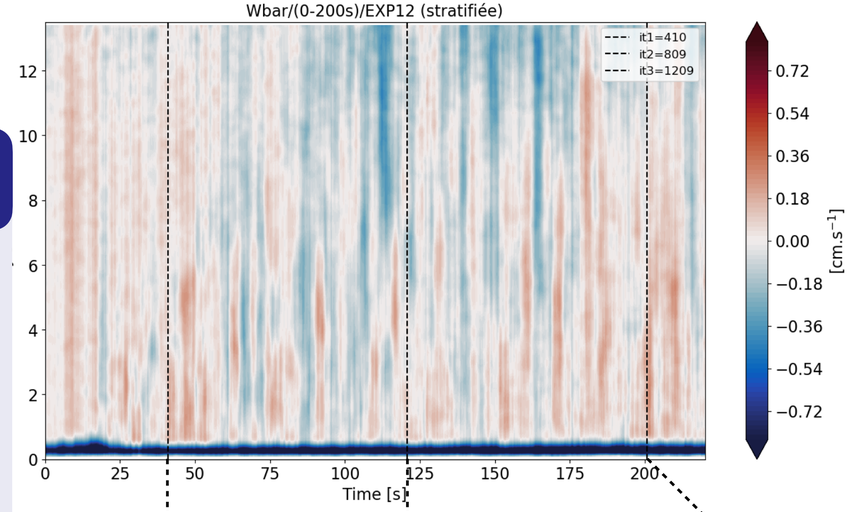
Profils verticaux de Wbar à différents instants



Forced Convection Experiment: Vertical profiles of velocity







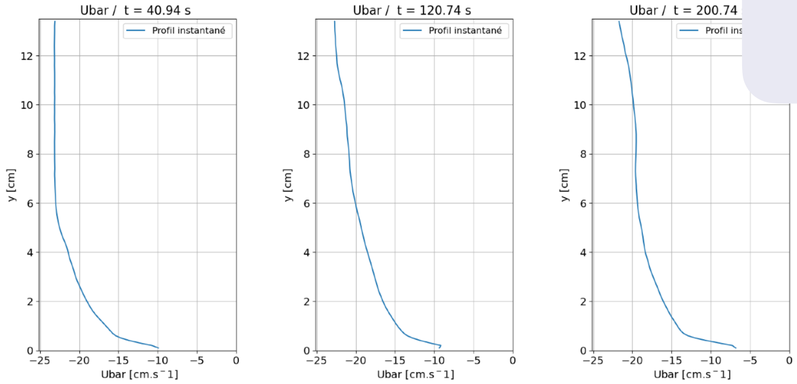
Momentum turbulent flux

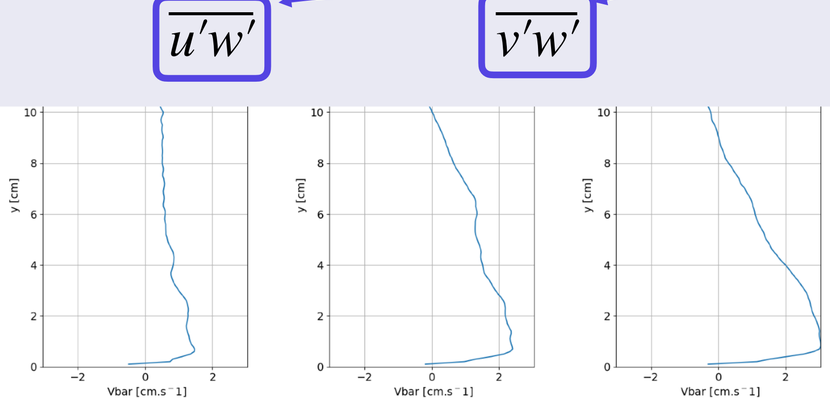
$$\mathbf{u} = u\,\mathbf{x_1} + v\,\mathbf{x_2} + w\,\mathbf{y}$$

$$u = \bar{u} + u' \qquad v = \bar{v} + v' \qquad w = \bar{w} + w'$$

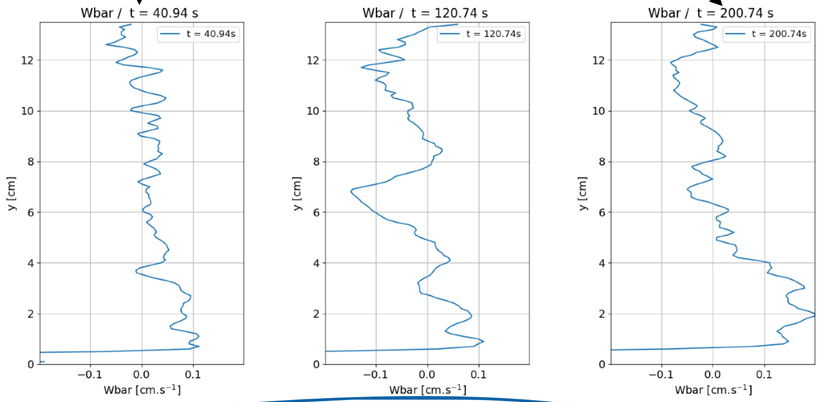
$$\overline{u'w'} \qquad \overline{v'w'}$$

Profils verticaux de Ubar à différents instants

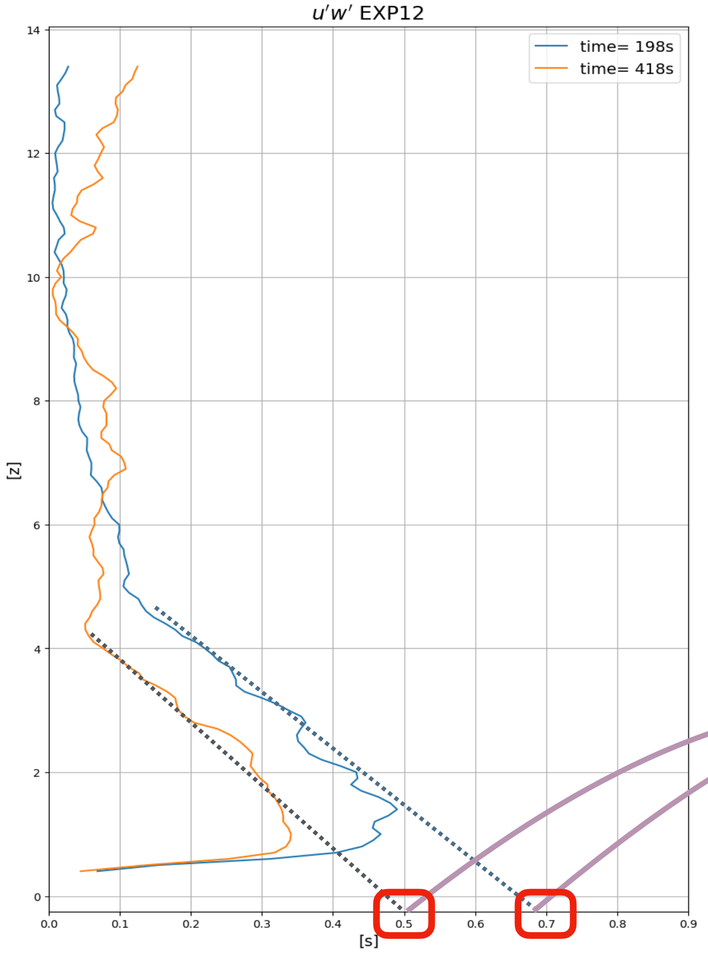




Profils verticaux de Wbar à différents instants



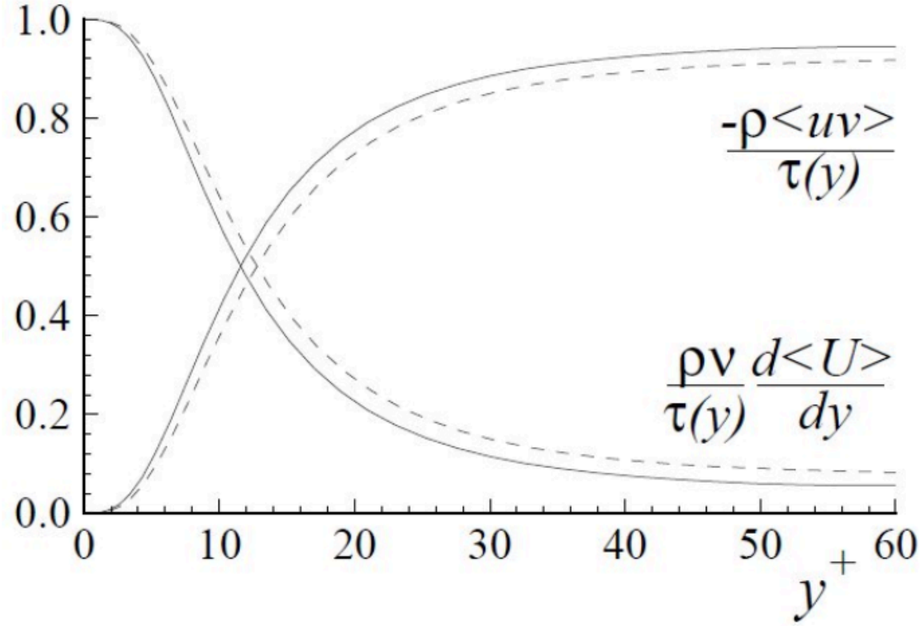
Forced Convection Experiment: Characterization of friction



Friction velocity

$$-u_*^2 = \overline{u'w'} - \nu\frac{\partial \overline{u}}{\partial z}$$

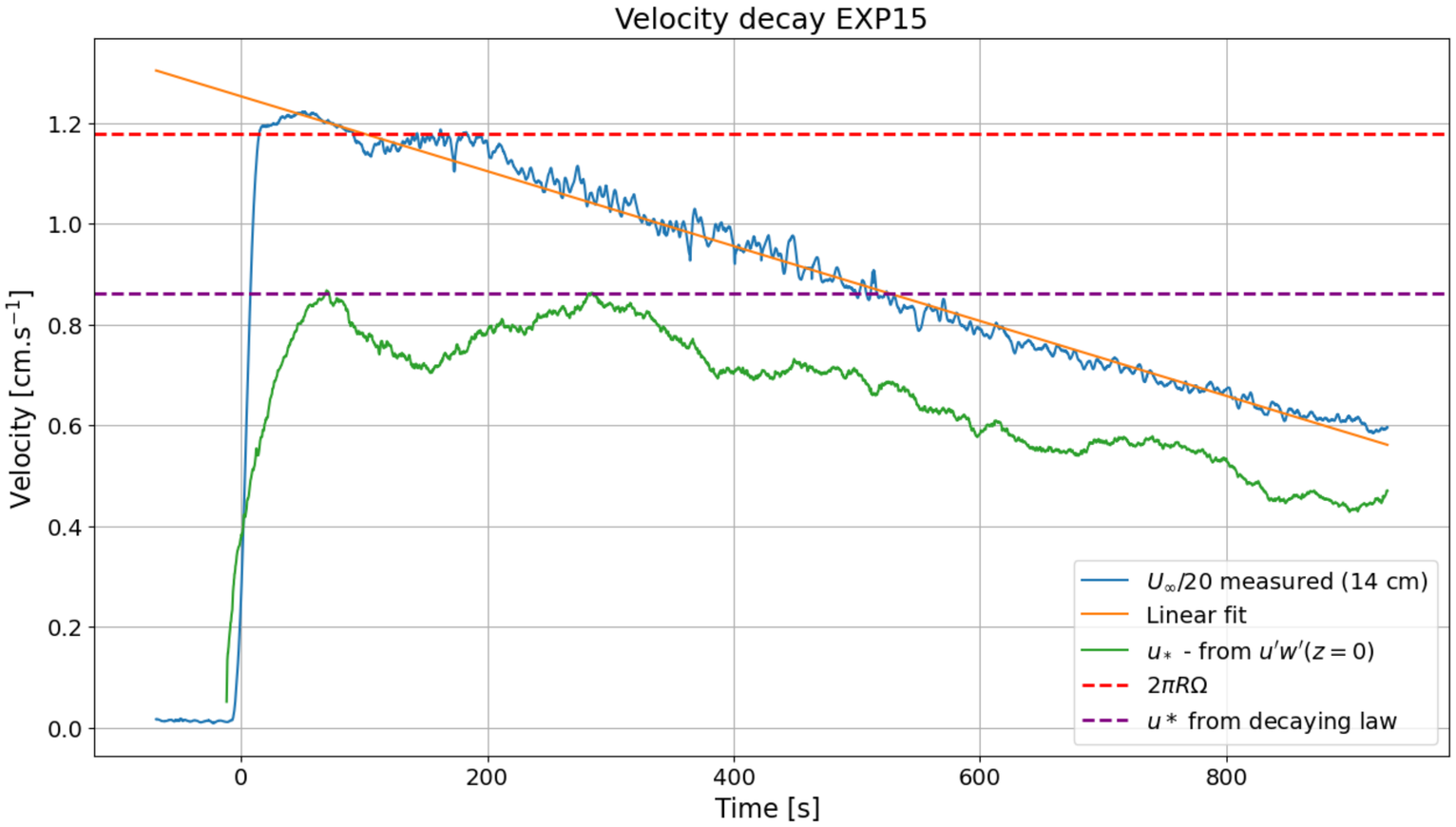
$$u_*^2(t)$$



Shear stress: turbulence + viscosity

Near the wall, the viscous and the turbulent stress sum up to a constant value (Pope, 2000)

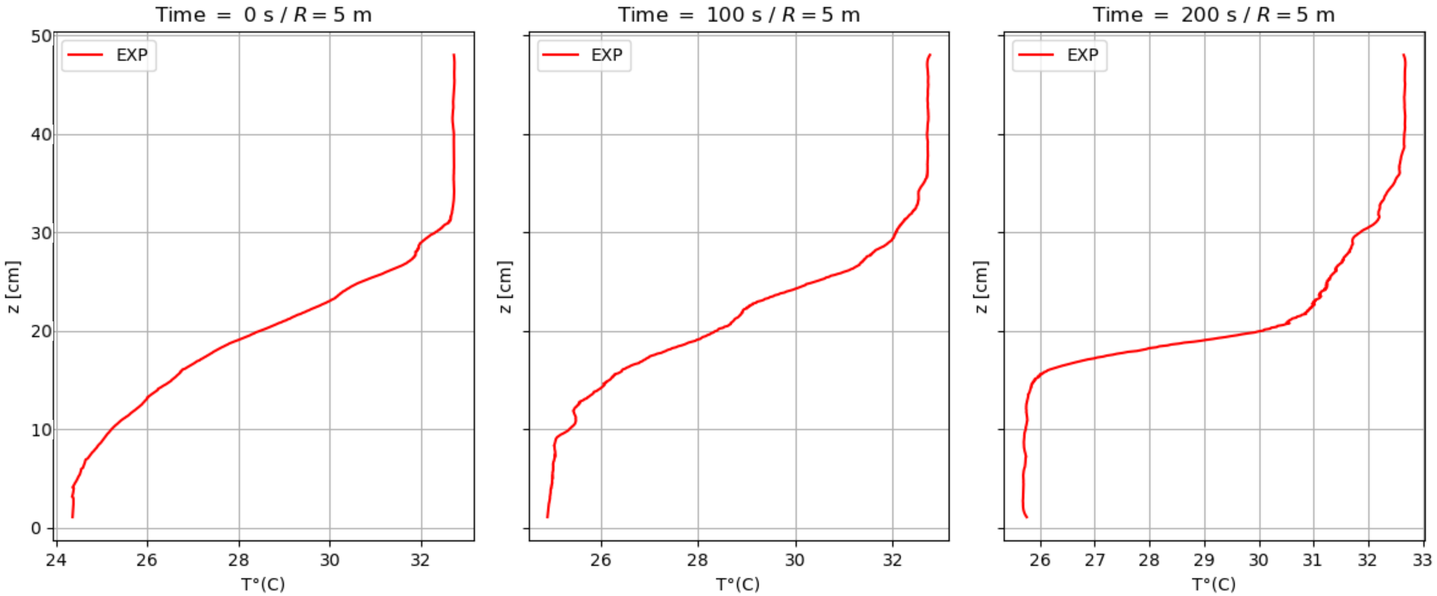
Forced convection experiment: Characterization of friction



Conservation of angular momentum

$$H\frac{dU}{dt} = -u_*^2$$

Forced Convection Experiment: Different Boundary layers

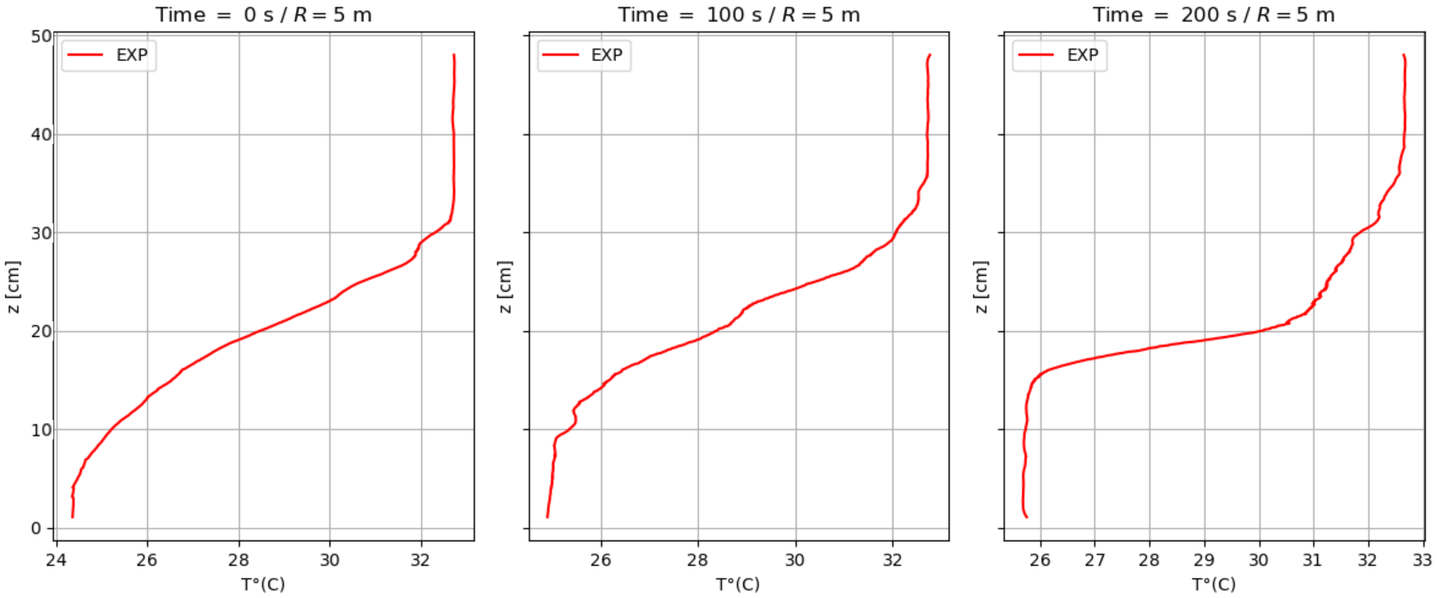


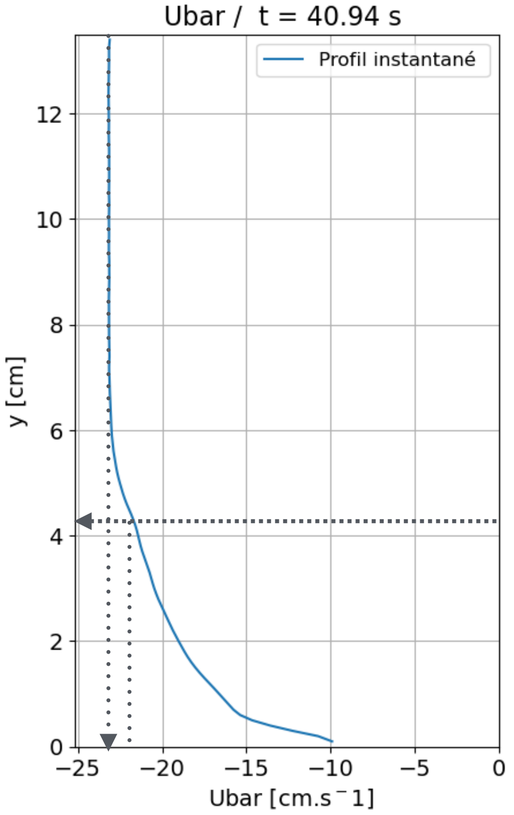
Boundary layer based on

Thermodynamical quantities

$$N^2(h_T) = \max(N^2)$$

Forced Convection Experiment: Different Boundary layers





Boundary layer based on Thermodynamical quantities

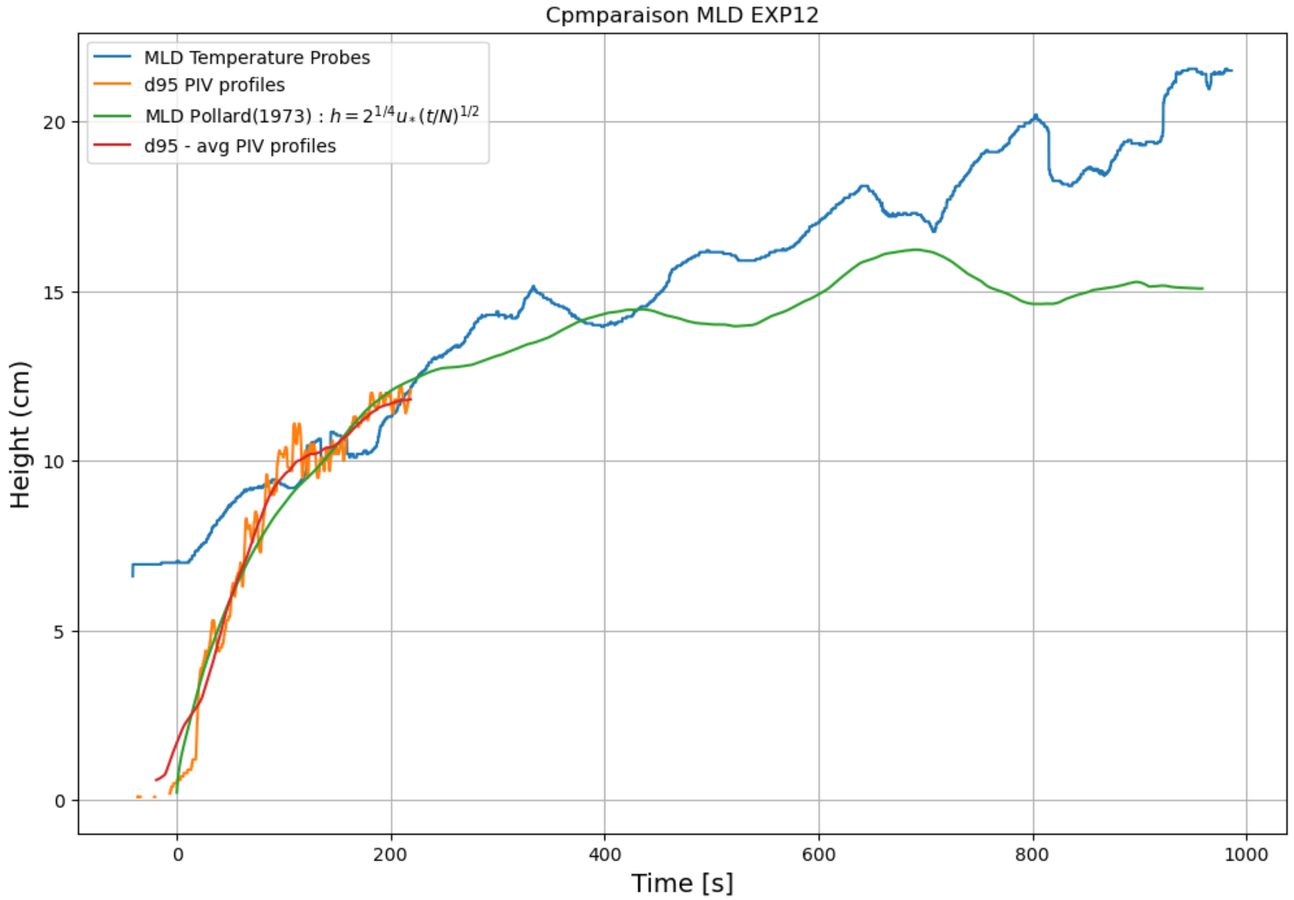
$$N^2(h_T) = \max(N^2)$$

Boundary layer based on dynamical quantities

$$U(\delta_{95}(t)) = 95\,\%\,U_\infty$$

With U_∞ the velocity of the fluid in the reference frame of the plate far from the wall

Forced Convection Experiment: Growth of the mixed layer



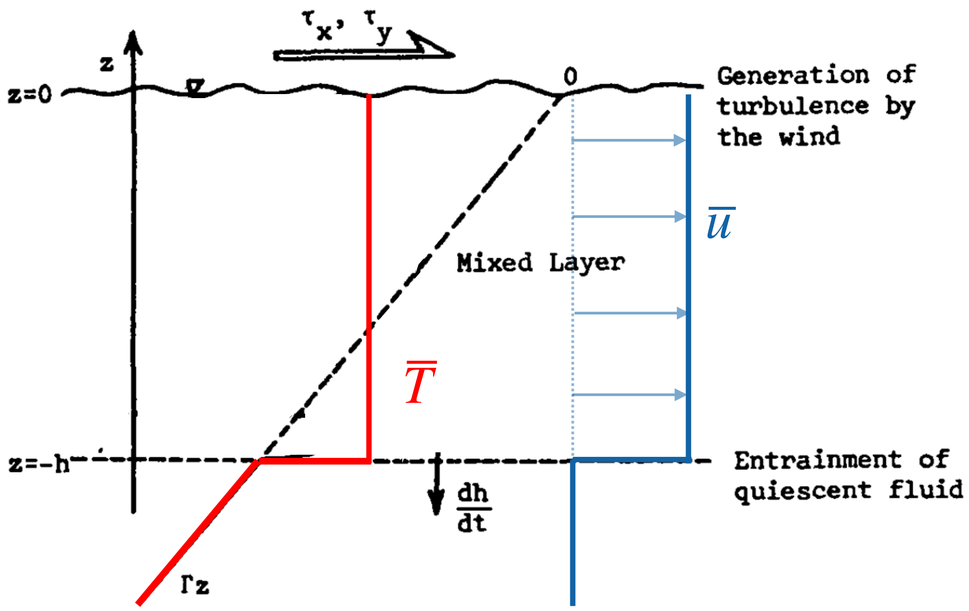
Growth of the mixed layer by momentum turbulent flux

- Accordance at short time with Pollard et al, 1973

$$h_{(t)} = 2^{1/4} u_* \left(\frac{t}{N} \right)^{1/2}$$

- Consistency between h_T and δ_{95} at short time
- At longer time $\longrightarrow$ different dynamics
  - Spin-up
  - Ekman dynamic

Longer time behavior of the ML: Slab model (Pollard et al, 1973)



- Uniform moving layer:

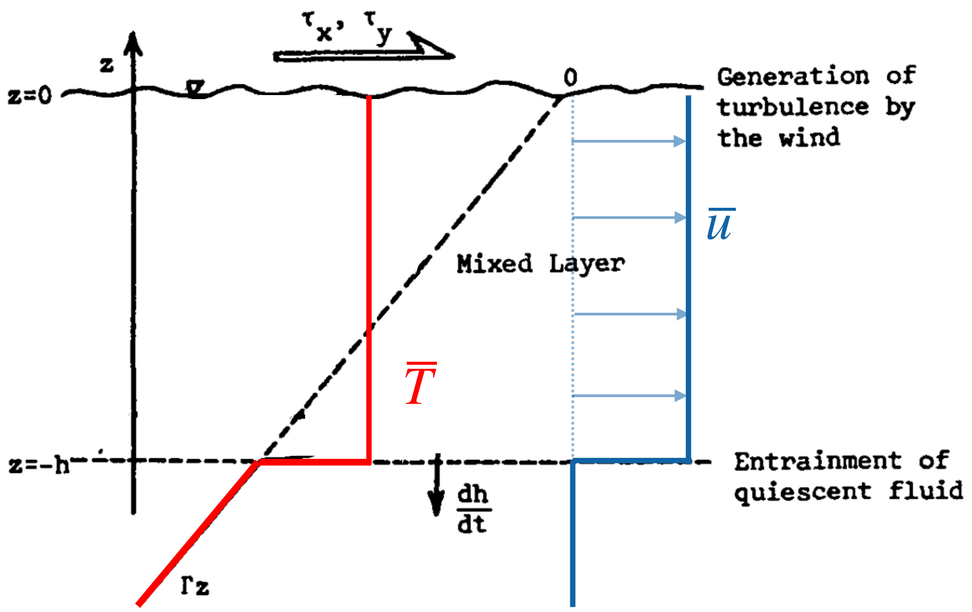
$$\frac{d\langle u\rangle}{dt} - f\langle v\rangle = u_*^2 \qquad \frac{d\langle v\rangle}{dt} + f\langle u\rangle = 0.$$

$$\langle u\rangle = \frac{u_*^2}{f}\sin(ft) \qquad \langle v\rangle = -\frac{u_*^2}{f}(1-\cos(ft))$$

$$E_{slab} = \left[\langle u\rangle^2 + \langle v\rangle^2\right]/2 \quad = \frac{u_*^4}{f^2}(1-\cos ft)$$

$$\frac{dE_{slab}}{dt} = u_*^2\ \langle u\rangle$$

Longer time behavior of the ML: Slab model (Pollard et al, 1973)



- Uniform moving layer:

$$\frac{d\langle u\rangle}{dt} - f\langle v\rangle = u_*^2 \qquad \frac{d\langle v\rangle}{dt} + f\langle u\rangle = 0.$$

$$\langle u\rangle = \frac{u_*^2}{f}\sin(ft) \qquad \langle v\rangle = -\frac{u_*^2}{f}(1-\cos(ft))$$

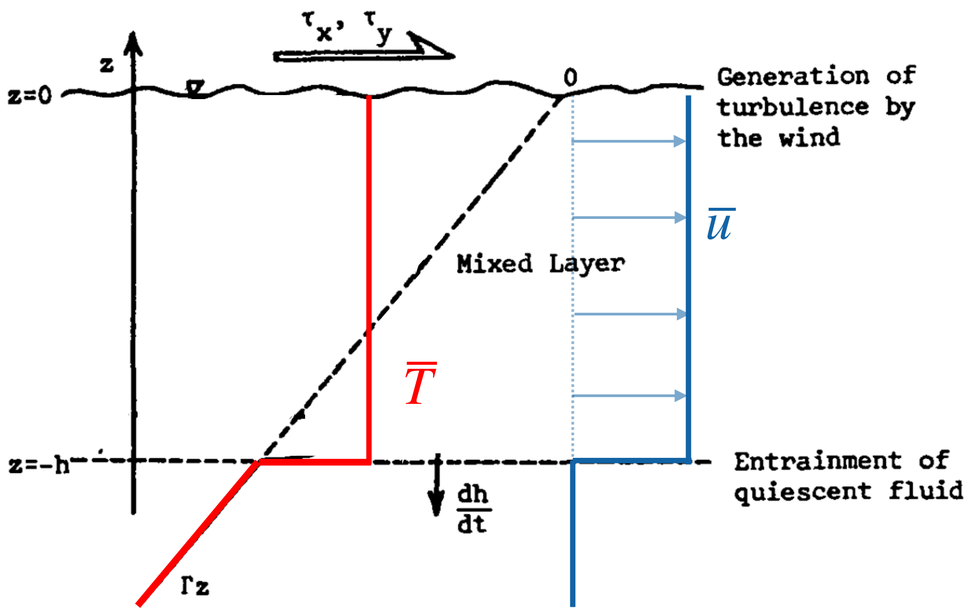
$$E_{slab} = \left[\langle u\rangle^2 + \langle v\rangle^2\right]/2 \quad = \frac{u_*^4}{f^2}(1-\cos ft)$$

$$\frac{dE_{slab}}{dt} = u_*^2\ \langle u\rangle$$

- Potential energy: (from geometrical considerations)

$$E_{pot} = -\int_{-H}^{0} bzdz = -\int_{-H}^{-h} N_0^2 z^2 dz + N_0^2\frac{h}{2}\int_{-h}^{0} zdz = N_0^2[\frac{h^3 - H^3}{3} - \frac{h^3}{4}]$$

Longer time behavior of the ML: Slab model (Pollard et al, 1973)



- Uniform moving layer:

$$\frac{d\langle u\rangle}{dt} - f\langle v\rangle = u_*^2 \qquad \frac{d\langle v\rangle}{dt} + f\langle u\rangle = 0.$$

$$\langle u\rangle = \frac{u_*^2}{f}\sin(ft) \qquad \langle v\rangle = -\frac{u_*^2}{f}(1-\cos(ft))$$

$$E_{slab} = \left[\langle u\rangle^2 + \langle v\rangle^2\right]/2 = \frac{u_*^4}{f^2}(1-\cos ft)$$

$$\frac{dE_{slab}}{dt} = u_*^2\ \langle u\rangle$$

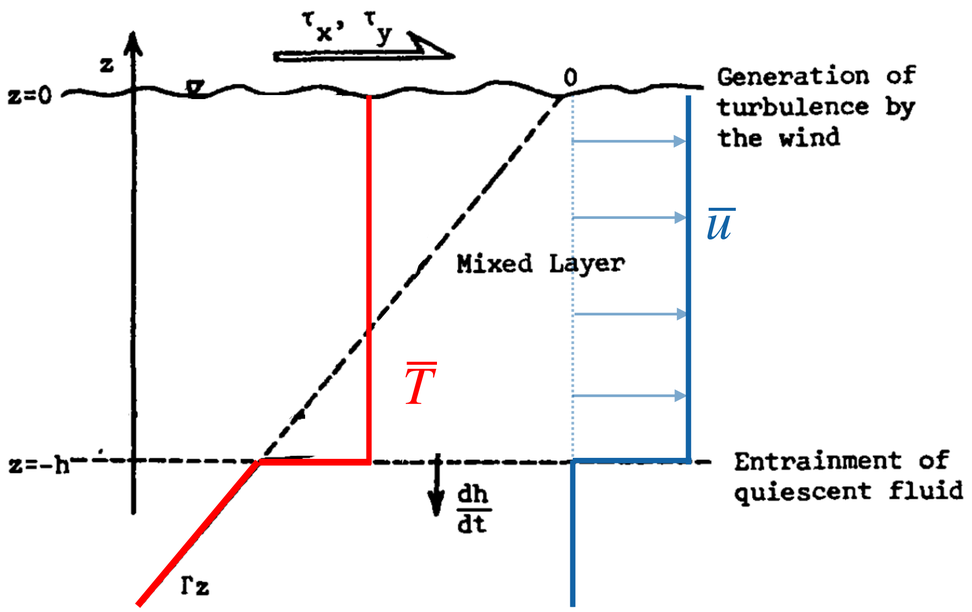
- Potential energy: (from geometrical considerations)

$$E_{pot} = -\int_{-H}^{0} bzdz = -\int_{-H}^{-h} N_0^2 z^2 dz + N_0^2\frac{h}{2}\int_{-h}^{0} zdz = N_0^2[\frac{h^3 - H^3}{3} - \frac{h^3}{4}]$$

- Kinetic energy: (Since $E_{kin} = E_{slab}/h$)

$$\frac{dE_{kin}}{dt} = \frac{dE_{slab}}{h\,dt} - \frac{dh}{h^2\,dt}E_{slab}$$

Longer time behavior of the ML: Slab model (Pollard et al, 1973)



- Uniform moving layer:

$$\frac{d\langle u\rangle}{dt} - f\langle v\rangle = u_*^2 \qquad \frac{d\langle v\rangle}{dt} + f\langle u\rangle = 0.$$

$$\langle u\rangle = \frac{u_*^2}{f}\sin(ft) \qquad \langle v\rangle = -\frac{u_*^2}{f}(1-\cos(ft))$$

$$E_{slab} = \left[\langle u\rangle^2 + \langle v\rangle^2\right]/2 = \frac{u_*^4}{f^2}(1-\cos ft)$$

$$\frac{dE_{slab}}{dt} = u_*^2\ \langle u\rangle$$

- Potential energy: (from geometrical considerations)

$$E_{pot} = -\int_{-H}^{0} bzdz = -\int_{-H}^{-h} N_0^2 z^2 dz + N_0^2\frac{h}{2}\int_{-h}^{0} zdz = N_0^2[\frac{h^3 - H^3}{3} - \frac{h^3}{4}]$$

- Kinetic energy: (Since $E_{kin} = E_{slab}/h$)

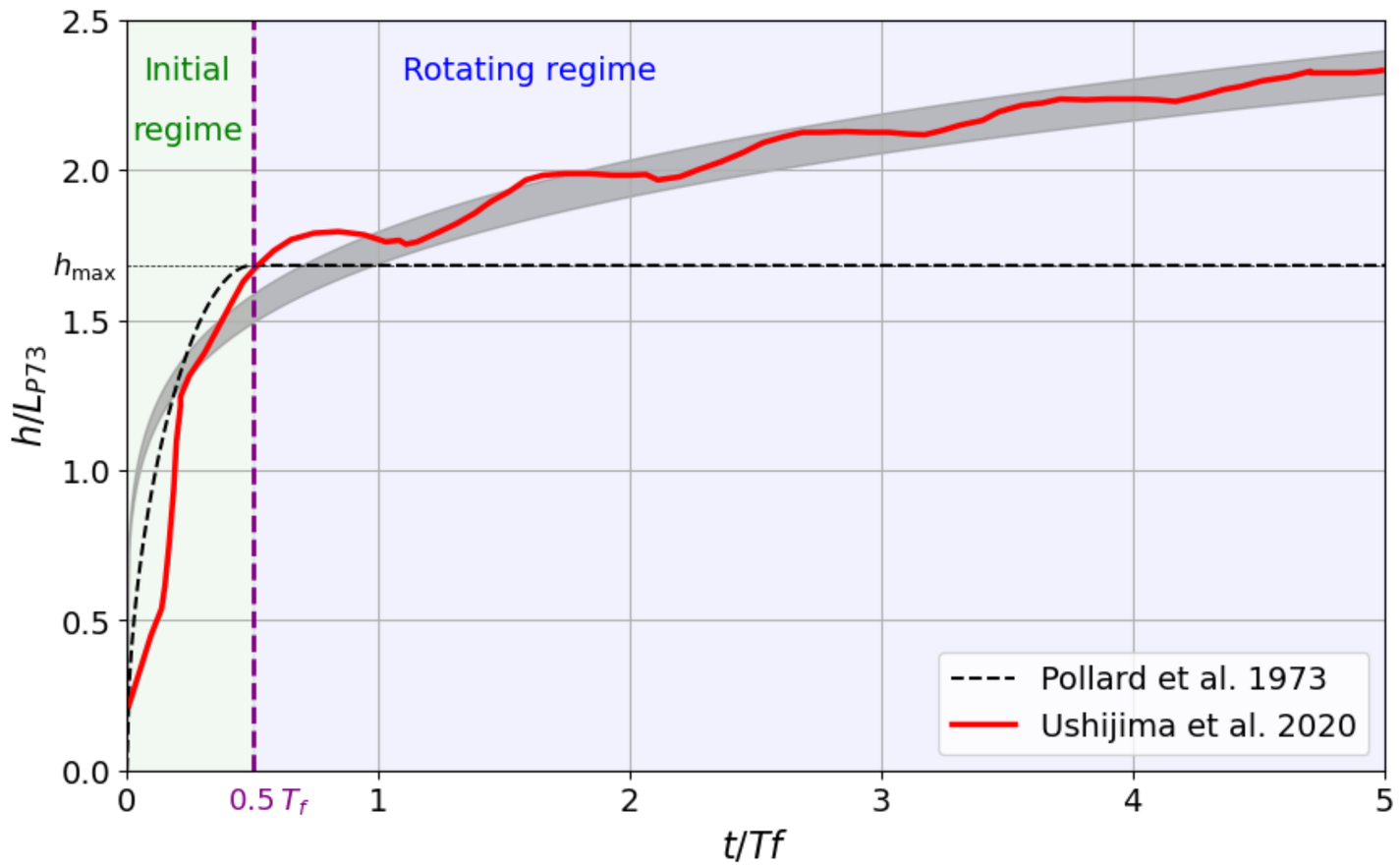
$$\frac{dE_{kin}}{dt} = \frac{dE_{slab}}{h\,dt} - \frac{dh}{h^2\,dt}E_{slab}$$

- Equation for energy:

$$\frac{dE_{kin}}{dt} + \frac{dE_{pot}}{dt} = u_*^2\ \langle u\rangle/h \qquad \rightarrow \qquad -E_{slab}\frac{dh}{h^2 dt} + \frac{dE_{pot}}{dt} = 0$$

$$\left[-\frac{u_*^4}{h^2 f^2}(1-\cos ft) + N_0^2\frac{h^2}{4}\right]\frac{dh}{dt} = 0 \qquad \rightarrow \qquad \frac{dh}{dt} = 0 \quad \text{or} \quad h^4 = \frac{4u_*^4}{f^2 N_0^2}(1-\cos ft)$$

Longer time behavior of the ML: Slab model (Pollard et al, 1973)



At longer time deepening continue

LES: $$h = 1.5 L_{p73} \left(\frac{N_0}{f}\right)^{0.022} \left(\frac{t}{T_f}\right)^{0.18}$$

Ushijima et al, 2020

Pollard et al, 1973 fail after $1/2\,T_f$

They consider at $t > \pi/f$ **that**

$$u_*^2 \langle u \rangle < 0$$

$$\left[-\frac{u_*^4}{h^2 f^2}(1-\cos ft) + N_0^2 \frac{h^2}{4}\right] \frac{dh}{dt} = 0$$

$$\frac{dh}{dt} = 0 \qquad \text{or} \qquad h^4 = \frac{4u_*^4}{f^2 N_0^2}(1-\cos ft)$$

Longer time behavior of the ML: Extended Slab model

- Slab energy: $\frac{dE_{slab}}{dt} = u_*^2 \ \langle u \rangle$
- Potential energy: $E_{pot} = \ N_0^2 [\frac{h^3 - H^3}{3} - \frac{h^3}{4}]$
- Kinetic energy: $E_{kin} = E_{slab}/h + u_*^2 \tilde{E}_{(h)}$

Longer time behavior of the ML: Extended Slab model

- Slab energy: $\frac{dE_{slab}}{dt} = u_*^2 \ \langle u \rangle$
- Potential energy: $E_{pot} = \ N_0^2[\frac{h^3 - H^3}{3} - \frac{h^3}{4}]$
- Kinetic energy: $E_{kin} = E_{slab}/h + u_*^2 \tilde{E}_{(h)}$

$\tilde{E}_{(h)}$ expresses the kinetic energy associated with the deviation of the velocity from the uniform slab velocity.

$$\tilde{E} = \frac{1}{2u_*^2} \int (\mathbf{u} - \frac{\langle \mathbf{u} \rangle}{h})^2 dz = \frac{1}{u_*^2} \left[E_{kin} - \frac{u_*^4}{f^2 h} (1 - cos(ft) \right]$$

Longer time behavior of the ML: Extended Slab model

- Slab energy: $\frac{dE_{slab}}{dt} = u_*^2 \ \langle u \rangle$
- Potential energy: $E_{pot} = \ N_0^2[\frac{h^3 - H^3}{3} - \frac{h^3}{4}]$
- Kinetic energy: $E_{kin} = E_{slab}/h + u_*^2 \tilde{E}_{(h)}$

$\tilde{E}_{(h)}$ expresses the kinetic energy associated with the deviation of the velocity from the uniform slab velocity.

$$\tilde{E} = \frac{1}{2u_*^2}\int(\mathbf{u} - \frac{\langle \mathbf{u} \rangle}{h})^2 dz = \frac{1}{u_*^2}\left[E_{kin} - \frac{u_*^4}{f^2 h}(1 - cos(ft)\right]$$

- Exact equation for energy:

$$\underbrace{\left[\underbrace{-\frac{u_*^2}{h^2 f^2}(1-\cos(ft))}_{\text{SLAB}} + \underbrace{\frac{N_0^2}{u_*^2}\frac{h^2}{4}}_{\text{POTENTIAL}}\right]\frac{dh}{dt}}_{>0} = \overbrace{\underbrace{\left(u_{(z=0,t)} - \frac{\langle u \rangle}{h}\right)}_{\text{EXTRA PROD}} - \underbrace{\frac{\mathcal{E}}{u_*^2}}_{\text{DISSIPATION}} + \underbrace{\frac{d}{dt}\tilde{E}}_{\text{DEVIATION}} + \underbrace{\frac{d}{dt}\tilde{E}_{turb}}_{\text{IMBALANCE}}}^{=\mathcal{R}}.$$

Longer time behavior of the ML: Extended Slab model

- Slab energy: $\frac{dE_{slab}}{dt} = u_*^2 \ \langle u \rangle$
- Potential energy: $E_{pot} = \ N_0^2[\frac{h^3 - H^3}{3} - \frac{h^3}{4}]$
- Kinetic energy: $E_{kin} = E_{slab}/h + u_*^2 \tilde{E}_{(h)}$

$\tilde{E}_{(h)}$ expresses the kinetic energy associated with the deviation of the velocity from the uniform slab velocity.

$$\tilde{E} = \frac{1}{2u_*^2} \int (\mathbf{u} - \frac{\langle \mathbf{u} \rangle}{h})^2 dz = \frac{1}{u_*^2} \left[E_{kin} - \frac{u_*^4}{f^2 h} (1 - cos(ft) \right]$$

- Exact equation for energy:

$$\underbrace{\left[\underbrace{-\frac{u_*^2}{h^2 f^2}(1 - \cos(ft))}_{\text{SLAB}} + \underbrace{\frac{N_0^2}{u_*^2}\frac{h^2}{4}}_{\text{POTENTIAL}} \right] \frac{dh}{dt}}_{> 0} = \overbrace{\underbrace{\left(u_{(z=0,t)} - \frac{\langle u \rangle}{h} \right)}_{\text{EXTRA PROD}} - \underbrace{\frac{\mathcal{E}}{u_*^2}}_{\text{DISSIPATION}} + \underbrace{\frac{d}{dt}\tilde{E}}_{\text{DEVIATION}} + \underbrace{\frac{d}{dt}\tilde{E}_{turb}}_{\text{IMBALANCE}}}^{=\mathcal{R}}$$

- Residual terms: $\mathscr{R} \simeq m_p \frac{u_*^4}{hf}$

$\mathscr{R}$ is a fraction m_p of the turbulent kinetic energy production in the entrainment layer

Longer time behavior of the ML: Extended Slab model

- Slab energy: $\frac{dE_{slab}}{dt} = u_*^2 \ \langle u \rangle$
- Potential energy: $E_{pot} = \ N_0^2[\frac{h^3 - H^3}{3} - \frac{h^3}{4}]$
- Kinetic energy: $E_{kin} = E_{slab}/h + u_*^2 \tilde{E}_{(h)}$

$\tilde{E}_{(h)}$ expresses the kinetic energy associated with the deviation of the velocity from the uniform slab velocity.

$$\tilde{E} = \frac{1}{2u_*^2}\int(\mathbf{u} - \frac{\langle \mathbf{u} \rangle}{h})^2 dz = \frac{1}{u_*^2}\left[E_{kin} - \frac{u_*^4}{f^2 h}(1 - cos(ft)\right]$$

- Exact equation for energy:

$$\underbrace{\left[\underbrace{-\frac{u_*^2}{h^2 f^2}(1-\cos(ft))}_{\text{SLAB}} + \underbrace{\frac{N_0^2}{u_*^2}\frac{h^2}{4}}_{\text{POTENTIAL}}\right]\frac{dh}{dt}}_{>0} = \overbrace{\underbrace{\left(u_{(z=0,t)} - \frac{\langle u \rangle}{h}\right)}_{\text{EXTRA PROD}} - \underbrace{\frac{\mathcal{E}}{u_*^2}}_{\text{DISSIPATION}} + \underbrace{\frac{d}{dt}\tilde{E}}_{\text{DEVIATION}} + \underbrace{\frac{d}{dt}\tilde{E}_{turb}}_{\text{IMBALANCE}}}^{=\mathcal{R}}.$$

- Residual terms: $\mathscr{R} \simeq m_p \frac{u_*^4}{hf}$

$\mathscr{R}$ is a fraction m_p of the turbulent kinetic energy production in the entrainment layer

$$\left[-\frac{u_*^4}{f^2 h^2}(1-\cos(ft)) + N_0^2\frac{h^2}{4}\right]\frac{dh}{dt} = m_p\frac{u_*^4}{hf}$$

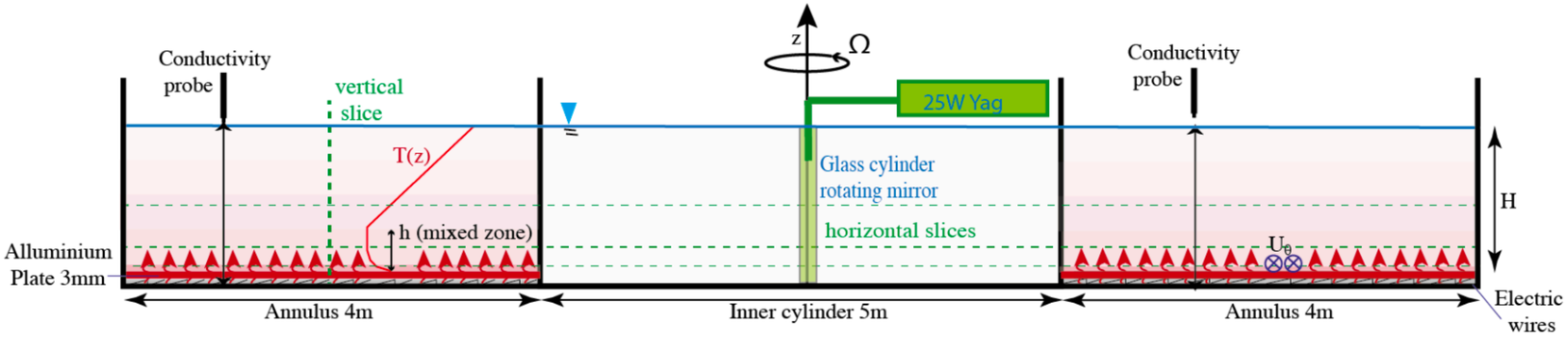
Consistant with scaling from LES

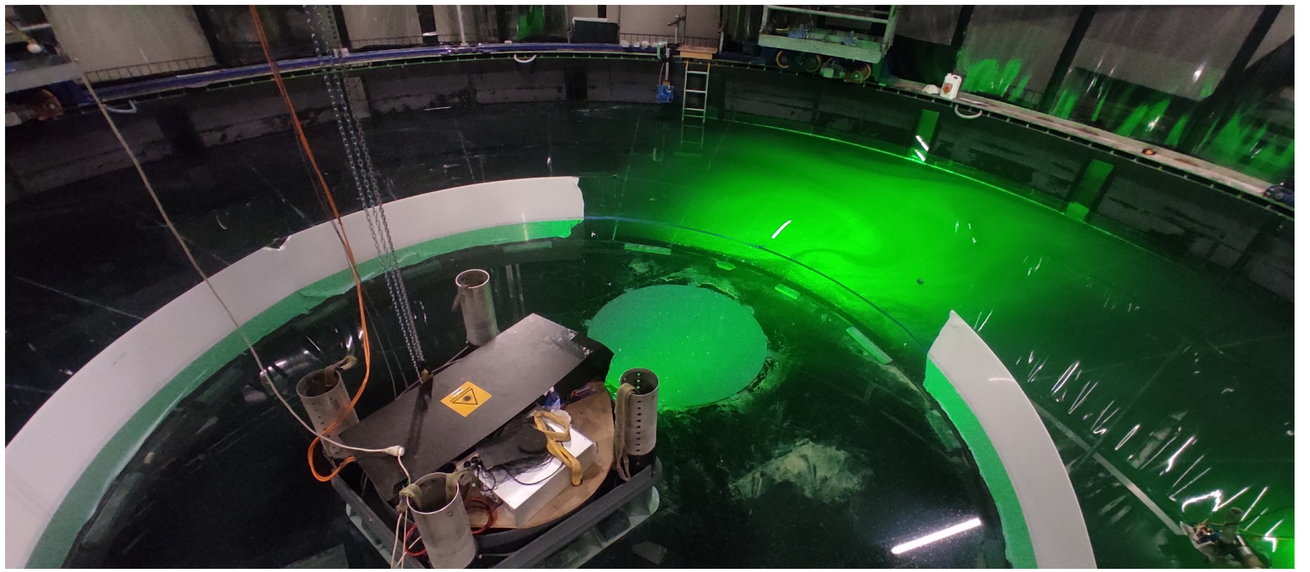
$$h^3\frac{dh}{dt} = 4m_p\frac{u_*^4}{N_0^2 f}$$

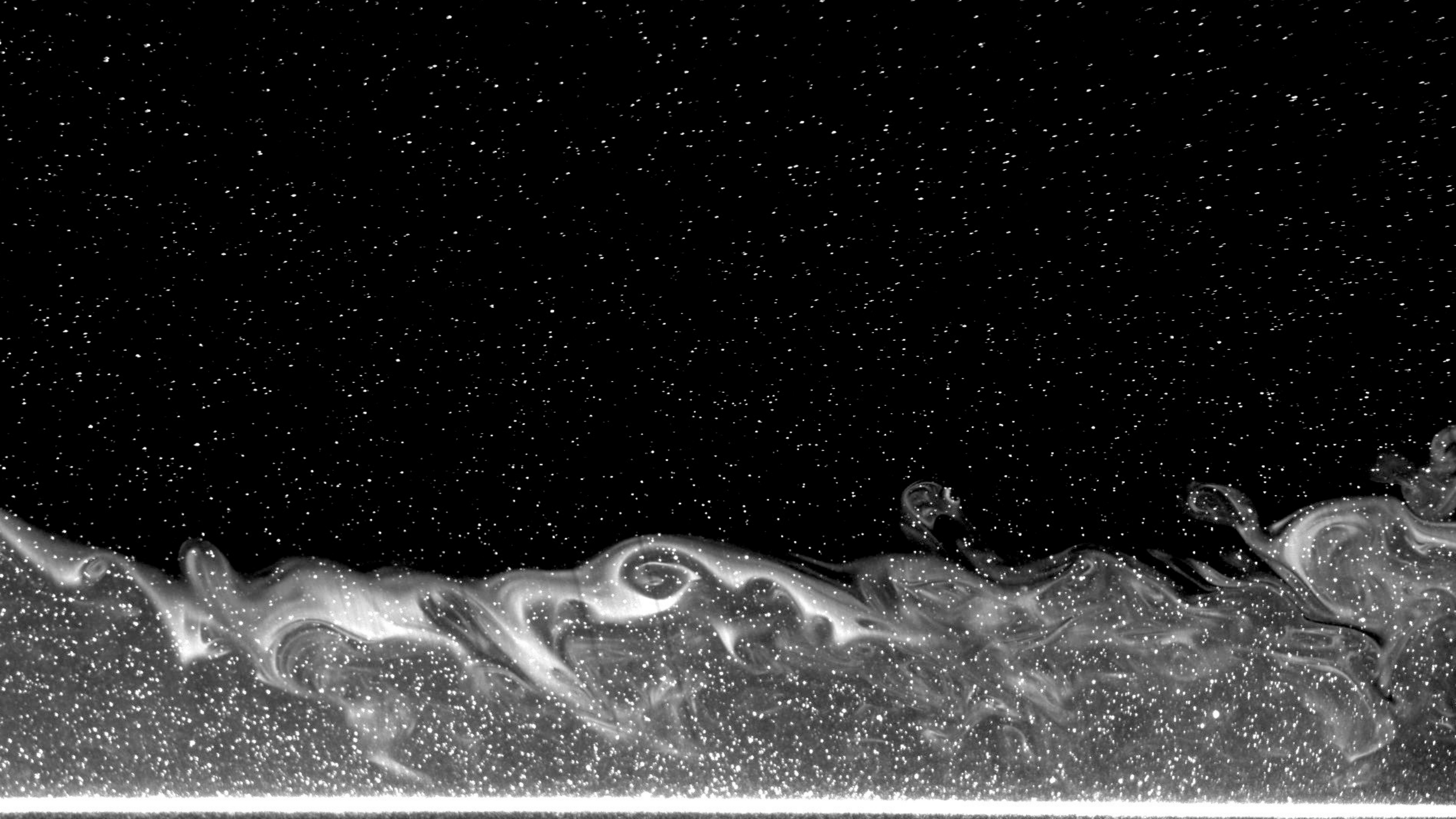
Paper in Prep

Next step : Free / Mixed Convection

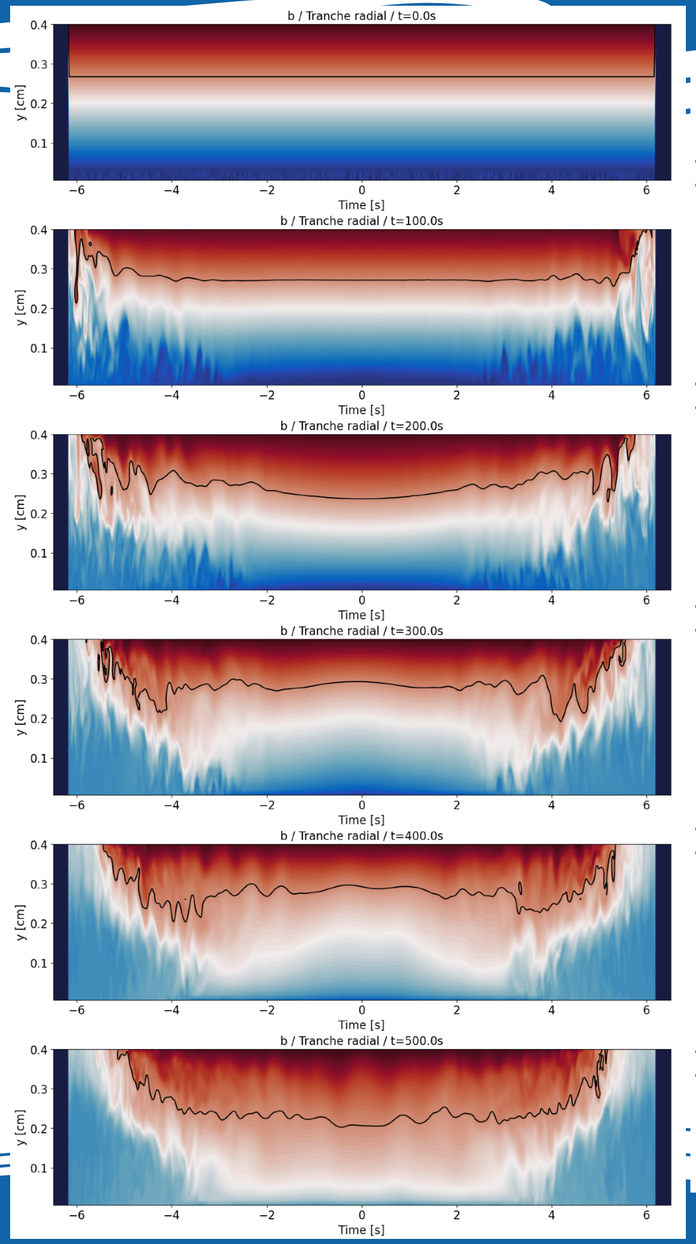
- **Heated floor [290-353] kW**
- **Inner cylinder (5m)**
- **Temperature probes**
  - 3 Vertical profilers
  - 2 Fixed probes (z =0; 12cm)
- **Vertical laser sheet (30x25)cm**
  - PIV Stereo
- **Horizontal laser sheet (3x4)m**
  - PIV (z = 10cm)
  - PIV in volume (multi- layer)
- **IR camera (3x4)m**

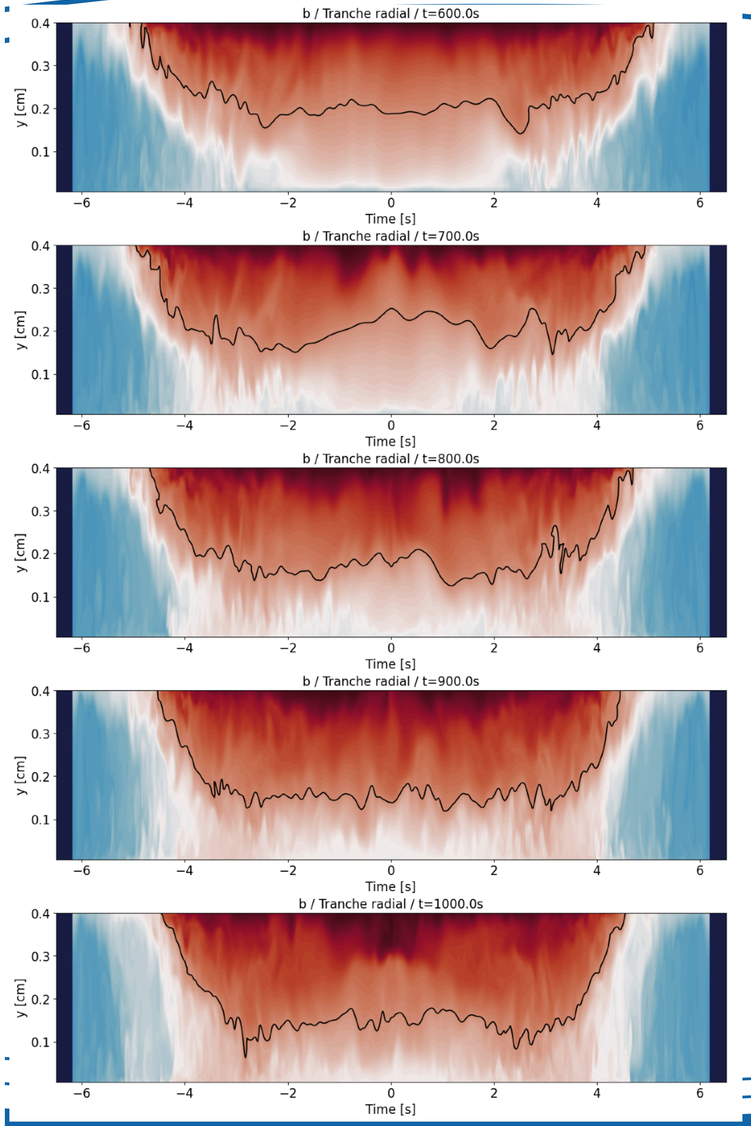






LES: Numerical Twins





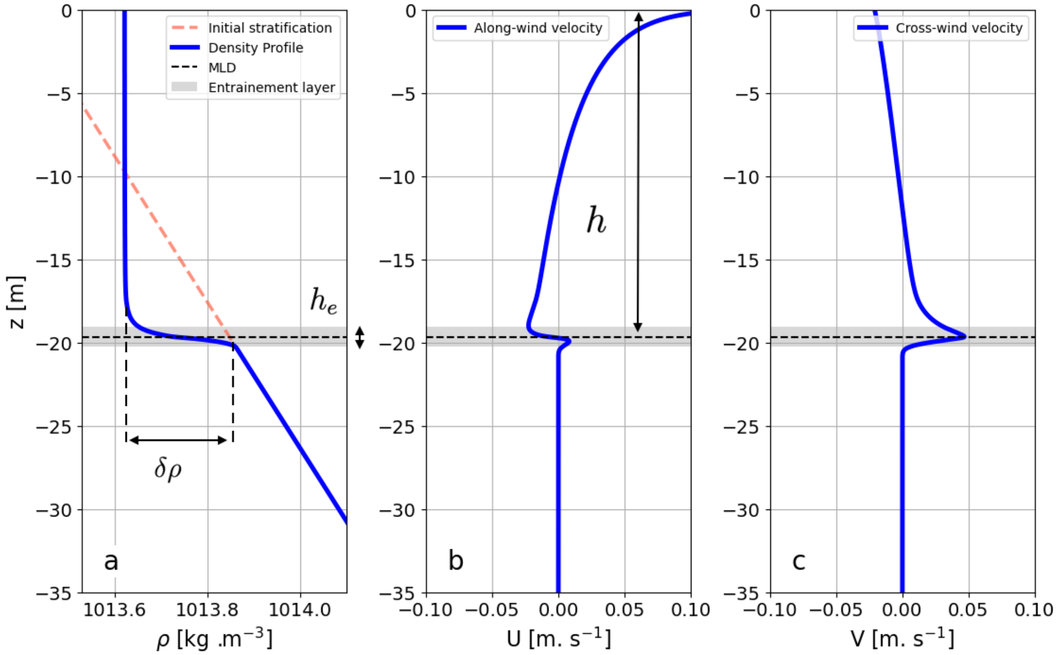
- At longer time different dynamics
  - **Strong radial effects**
  - **Wavy dynamics**

Basilics

LES - 1024 x 1024 x 32



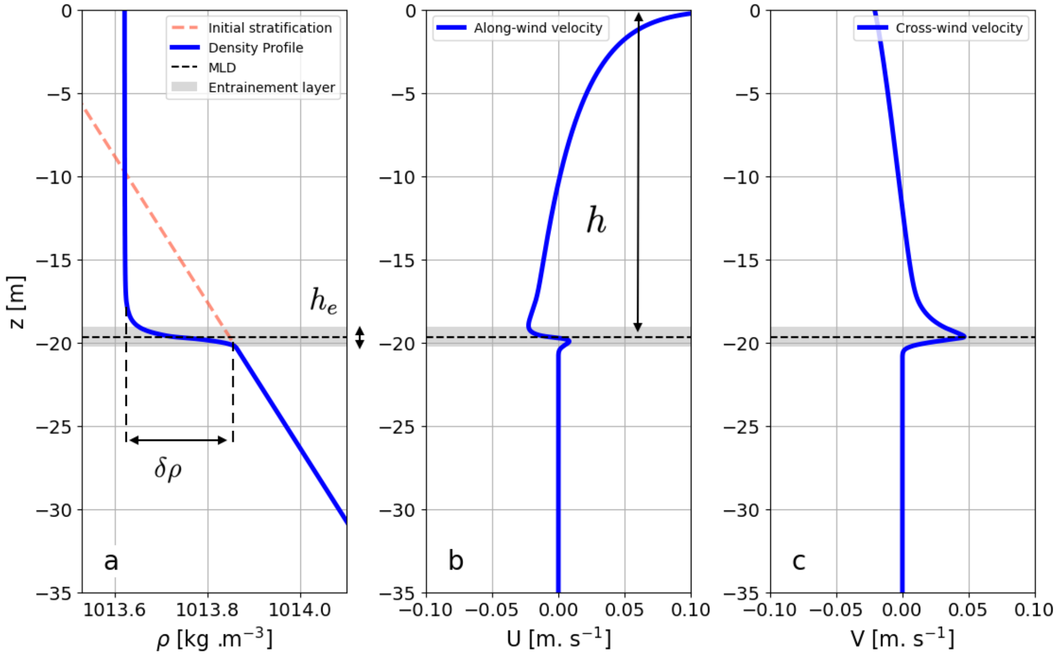
Longer time behavior of the ML: Entrainment layer model



- Considering a fully mixed layer and an entrainment layer h_e

  Mixing occurs only in the entrainement layer ⟶ Marginal stability

Longer time behavior of the ML: Entrainment layer model



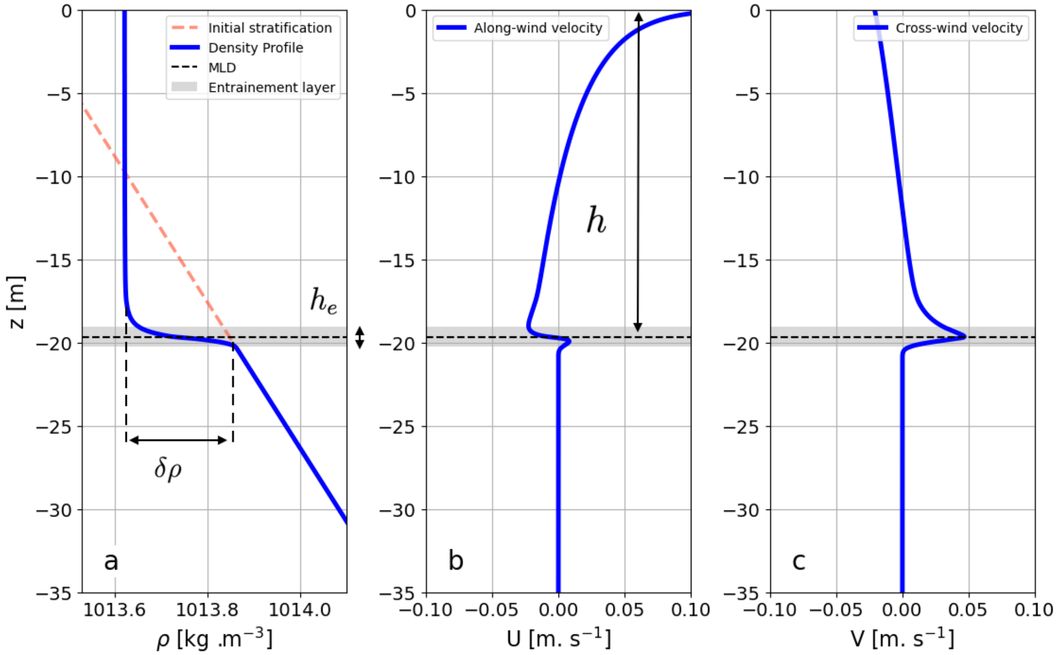
- Considering a fully mixed layer and an entrainment layer h_e

  Mixing occurs only in the entrainement layer ⟶ Marginal stability

  Buoyancy drop
  $\partial b/\partial z \simeq N_0^2 h/(2h_e)$

  Velocity drop (Averaged over T_f)
  $(\partial u/\partial z)^2 \simeq 2u_*^4/(f^2 h_e^2)$

Longer time behavior of the ML: Entrainment layer model



- Considering a fully mixed layer and an entrainment layer h_e

  Mixing occurs only in the entrainement layer ⟶ Marginal stability

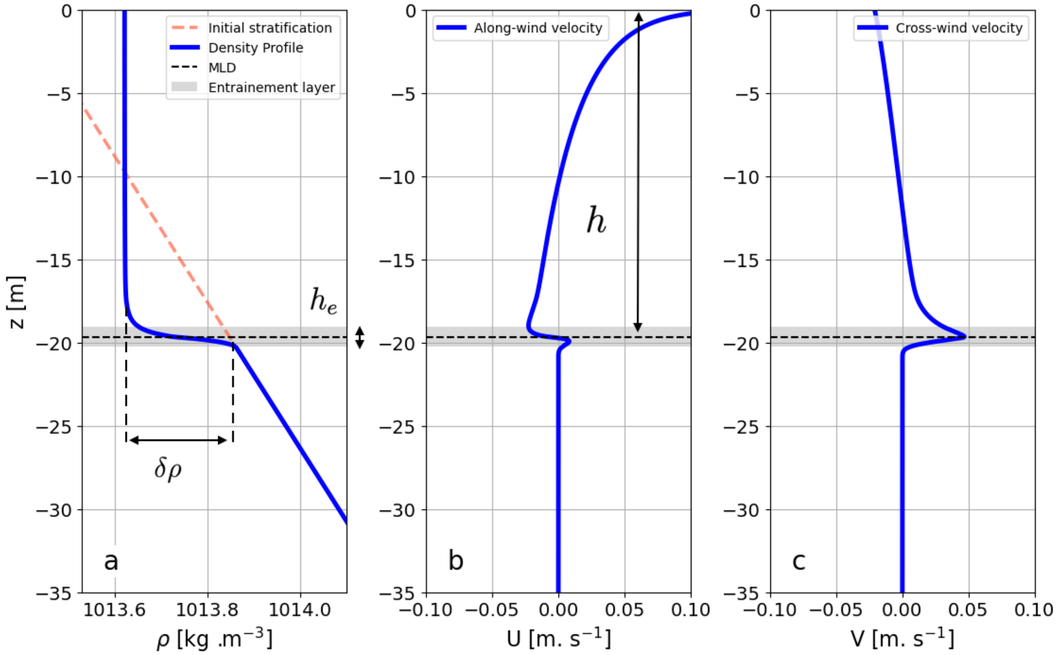
  Buoyancy drop
  $\partial b/\partial z \simeq N_0^2 h/(2h_e)$

  Velocity drop (Averaged over T_f)
  $(\partial u/\partial z)^2 \simeq 2u_*^4/(f^2 h_e^2)$

- Relation Between active/entrainment layer thickness:

$$\frac{h_e}{h} = 4Ri_{st}\frac{u_*^4}{h^4 N_0^2 f^2}$$

Longer time behavior of the ML: Entrainment layer model



- Considering a fully mixed layer and an entrainment layer h_e

  Mixing occurs only in the entrainement layer ⟶ Marginal stability

  Buoyancy drop
  $\partial b/\partial z \simeq N_0^2 h/(2h_e)$

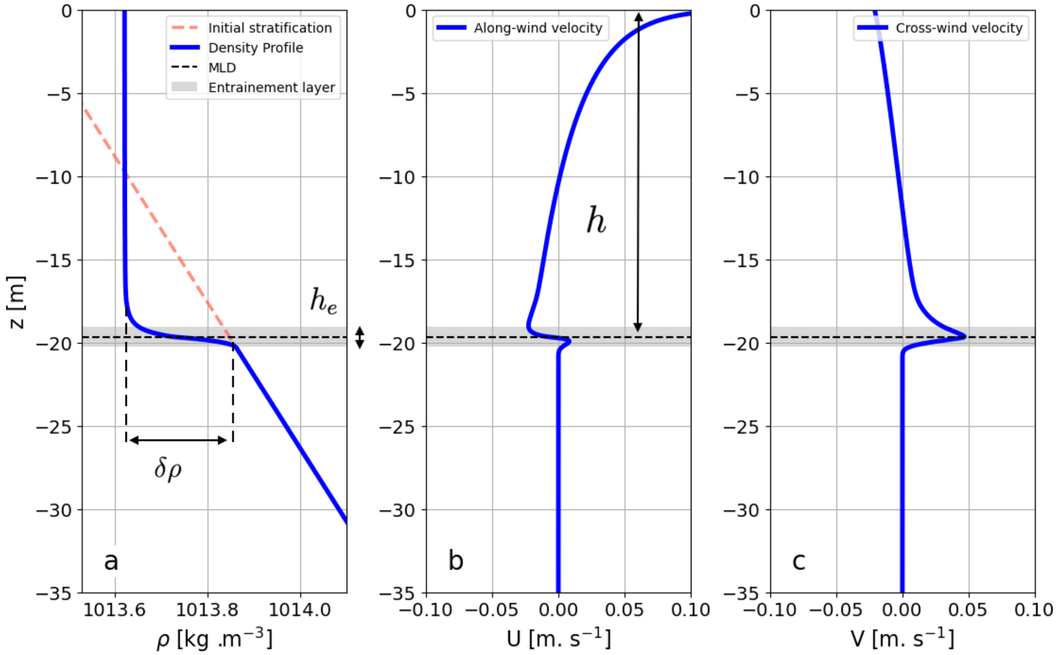
  Velocity drop (Averaged over T_f)
  $(\partial u/\partial z)^2 \simeq 2u_*^4/(f^2 h_e^2)$

- Relation Between active/entrainment layer thickness:

$$\frac{h_e}{h} = 4Ri_{st}\frac{u_*^4}{h^4 N_0^2 f^2}$$

- Local Buoyancy: $N_e^2 = N_0^2 h/(2h_e)$

Longer time behavior of the ML: Entrainment layer model



- Considering a fully mixed layer and an entrainment layer h_e

  Mixing occurs only in the entrainement layer ⟶ Marginal stability

  Buoyancy drop $\partial b/\partial z \simeq N_0^2 h/(2h_e)$

  Velocity drop (Averaged over T_f) $(\partial u/\partial z)^2 \simeq 2u_*^4/(f^2 h_e^2)$

- Relation Between active/entrainment layer thickness:

$$\frac{h_e}{h} = 4Ri_{st}\frac{u_*^4}{h^4 N_0^2 f^2}$$

- Local Buoyancy: $N_e^2 = N_0^2 h/(2h_e)$

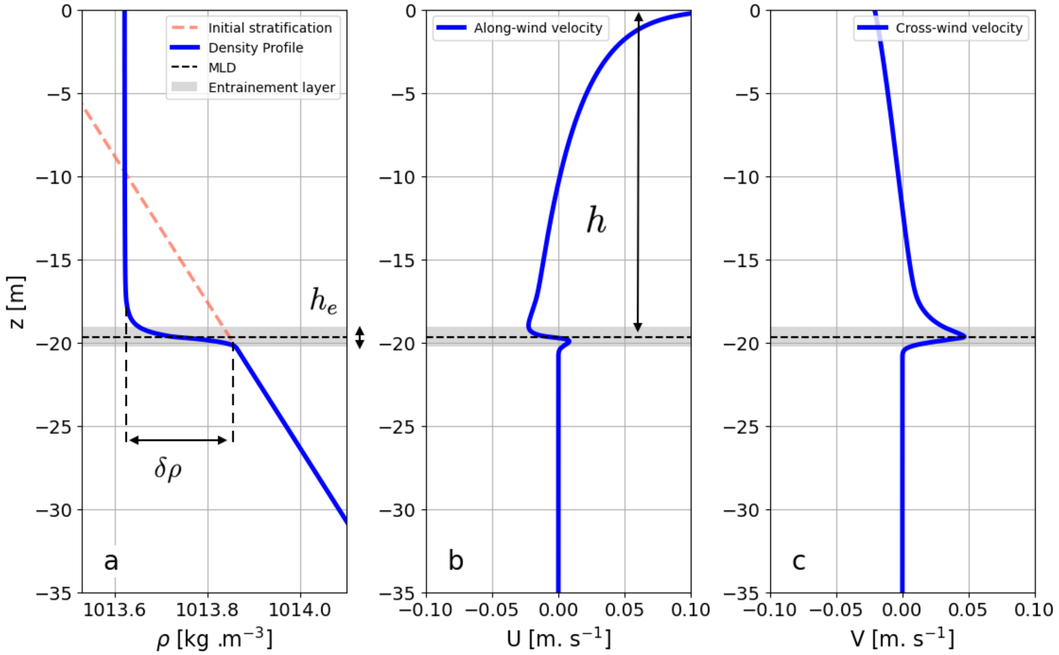
- Turbulent Viscosity: (From dimensional argument)

$$\nu_t \simeq u_*^2/N_e \simeq 2\sqrt{2}Ri_{st}^{1/2}\frac{u_*^4}{h^2 N_0^2 f}$$

- TKE Production: (integrated over the layer) $< P >_e \simeq \nu_t u^2/h_e$

$$u^2 \simeq 2u_*^4/(f^2 h^2)$$

Longer time behavior of the ML: Entrainment layer model



- Considering a fully mixed layer and an entrainment layer h_e

  Mixing occurs only in the entrainement layer ⟶ Marginal stability

  Buoyancy drop $\partial b/\partial z \simeq N_0^2 h/(2h_e)$

  Velocity drop (Averaged over T_f) $(\partial u/\partial z)^2 \simeq 2u_*^4/(f^2 h_e^2)$

- Relation Between active/entrainment layer thickness:

$$\frac{h_e}{h} = 4Ri_{st}\frac{u_*^4}{h^4 N_0^2 f^2}$$

- Local Buoyancy: $N_e^2 = N_0^2 h/(2h_e)$

- Turbulent Viscosity: (From dimensional argument)

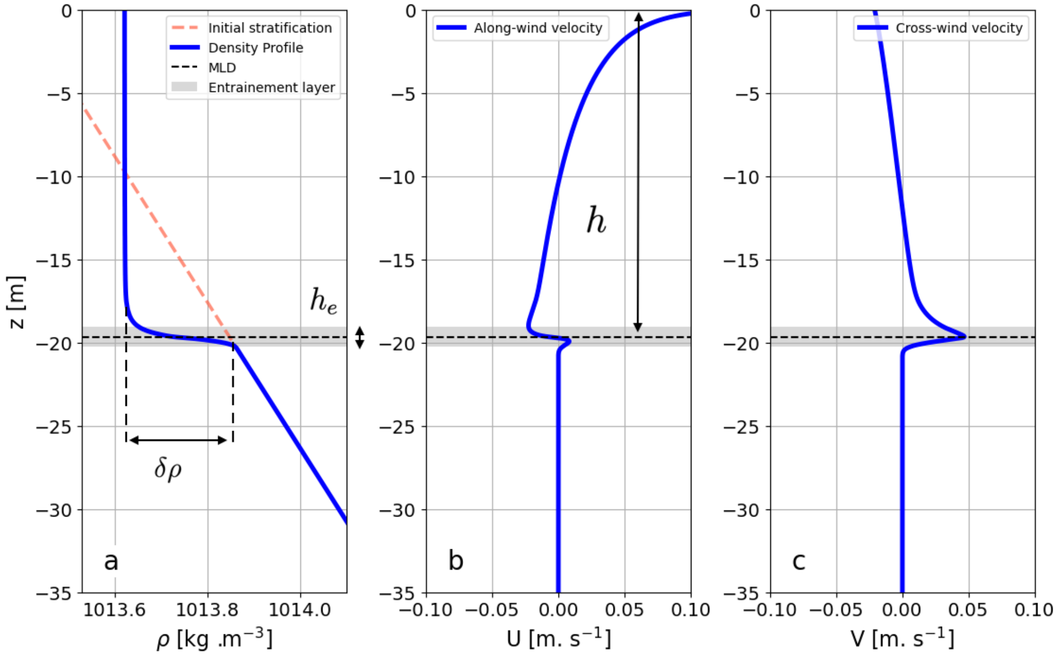
$$\nu_t \simeq u_*^2/N_e \simeq 2\sqrt{2}Ri_{st}^{1/2}\frac{u_*^4}{h^2 N_0^2 f}$$

- TKE Production: (integrated over the layer) $< P >_e \simeq \nu_t u^2/h_e$

  $u^2 \simeq 2u_*^4/(f^2 h^2)$

- Mixing efficiency $dE_p/dt \propto \frac{u_*^4}{hf} m_p < P >_e$

Longer time behavior of the ML: Entrainment layer model



$$\left[-\frac{u_*^4}{f^2h^2}(1-\cos(ft))+N_0^2\frac{h^2}{4}\right]\frac{dh}{dt}=m_p\frac{u_*^4}{hf}$$

$$h^3\frac{dh}{dt}=4m_p\frac{u_*^4}{N_0^2f}$$

- Considering a fully mixed layer and an entrainment layer h_e

  Mixing occurs only in the entrainement layer ⟶ Marginal stability

  Buoyancy drop $\partial b/\partial z \simeq N_0^2 h/(2h_e)$

  Velocity drop (Averaged over T_f) $(\partial u/\partial z)^2 \simeq 2u_*^4/(f^2h_e^2)$

- Relation Between active/entrainment layer thickness:

$$\frac{h_e}{h}=4Ri_{st}\frac{u_*^4}{h^4N_0^2f^2}$$

- Local Buoyancy: $N_e^2=N_0^2h/(2h_e)$

- Turbulent Viscosity: (From dimensional argument)

$$\nu_t \simeq u_*^2/N_e \simeq 2\sqrt{2}Ri_{st}^{1/2}\frac{u_*^4}{h^2N_0^2f}$$

- TKE Production: (integrated over the layer) $< P >_e \simeq \nu_t u^2/h_e$

  $u^2 \simeq 2u_*^4/(f^2h^2)$

- Mixing efficiency $dE_p/dt \propto \frac{u_*^4}{hf}m_p < P >_e$